

Geometry-Aware Neural Operator Surrogates for External Aerodynamics: Physics Consistency, Out-of-Distribution Generalization and Calibrated Uncertainty

T. Amin*

(Dated: September 9, 2026)

Geometry-conditioned neural surrogates are reported as accurate on external aerodynamics benchmarks, but almost always in one narrow sense: field or drag error, in distribution, with no check on whether a prediction is internally consistent or the reported uncertainty means anything. We audit three geometry-conditioning families (a k -nearest-neighbour graph network, a signed-distance-conditioned Fourier neural operator in the GINO style, and a Transolver-style geometry transformer), each reimplemented at matched parameter count (~ 1.3 – 1.4 M) and training budget, under a single protocol built around force self-consistency, generalization across six evaluation splits, and calibrated uncertainty. The central instrument is force self-consistency (FSC): the disagreement between the drag obtained by quadrature of a model’s own predicted surface pressure and the drag from a regression head on the same backbone. FSC needs no ground truth, so it is computable out of distribution and on unlabelled candidate geometries. On AirfRANS, with the force integrator gated against the dataset’s own coefficients (median relative error 3.3×10^{-4}), three findings recur across all three architectures. Field accuracy separates the models cleanly (transformer > operator > graph net, everywhere, by up to $4\times$), but C_D rank correlation from the coefficient head does not: all three sit at $\rho_s \approx 0.94$ – 0.999 , and a 29-parameter ridge baseline reaches 0.83 – 0.90 on every split. Ranking drag by integrating the predicted fields is weaker still, no better than ridge for the best model and far worse for the graph net, so the two drags a single model reports disagree by 5 – 24% in distribution and up to 115% under shift, which is exactly what makes FSC a working label-free shift detector. Split conformal prediction holds near nominal coverage in distribution for two of the three models but its guarantee does not survive flow-regime shift, and an uncertainty-driven acquisition score, verified against independent Ansys Fluent RANS runs, concentrates its picks in the corner of the design space where the steady solver itself struggles; on the two post-stall cases the solver could settle, the surrogates were off by 640 – $900\times$ their in-distribution error. Code, split manifests, and the Fluent verification set are released.

I. INTRODUCTION

The external-aerodynamics surrogate literature has converged on a narrow scoreboard: relative L_2 error on velocity and pressure fields, plus an error or rank correlation on drag. Papers get accepted on that scoreboard. Engineers cannot use it, for three reasons.

Gap 1: internal inconsistency is invisible. Models are trained with a field loss and, sometimes, a separate head for integrated coefficients. Nothing forces the predicted surface pressure and wall shear stress to integrate to the predicted drag. A model can be accurate in field norm while being internally inconsistent about the one number (an integrated force coefficient) a designer actually acts on, because force is a small difference of large, partially cancelling contributions dominated by regions (leading-edge suction peak, separation onset, base pressure) that carry little weight in a domain-averaged norm. That internal gap is a cheap, ground-truth-free diagnostic almost nobody reports; we make it a first-class measurement, and it turns out to be large: tens of percent of the true drag, growing with shift (§VIC).

Gap 2: out-of-distribution failure is acknowledged but not characterized. Reported gains are almost always measured on random, in-distribution splits, while deployment means new operating points and new shapes. Prior benchmarking work states plainly that OOD generalization remains a challenge for every method tested [1], which is a conclusion, not a characterization. Which axis of shift hurts most: Reynolds number, angle of attack, or shape family? We evaluate on six splits spanning four shift axes (data scarcity, Reynolds number, angle of attack, shape family) plus a fifth split that shifts shape and Reynolds together, the hardest cell, and report every metric against split identity rather than as a single aggregate (§III).

Gap 3: uncertainty is reported without calibration. Where uncertainty is reported at all it is usually an ensemble spread: a heuristic score, least trustworthy under exactly the shift where it matters, rarely checked against realised coverage and almost never checked *under shift*. Split conformal prediction converts any such score into finite-

* taimooramin419@gmail.com

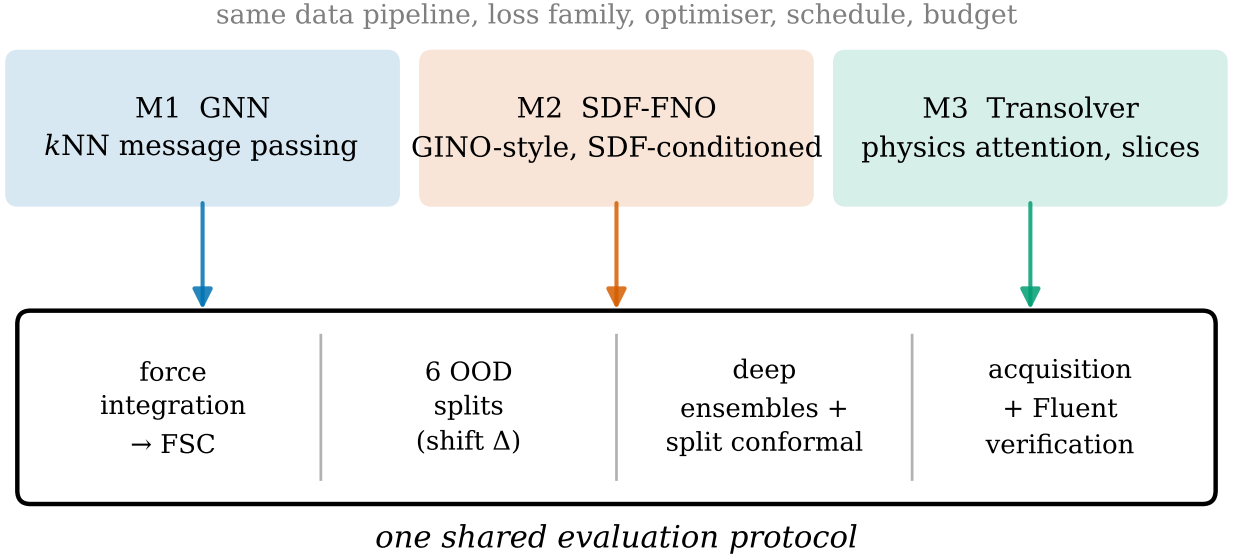


FIG. 1. The three geometry-encoding families and the shared evaluation protocol beneath them: only the encoder changes; the force integration, the six OOD splits, the ensembling/conformal calibration and the acquisition/Fluent loop are identical across M1-GNN, M2-SDF-FNO and M3-TRANSOLVER.

sample coverage under exchangeability, a genuine guarantee in distribution and a measurable failure mode out of it, yet conformal coverage for aerodynamic surface fields and derived coefficients has not been systematically reported.

a. Contributions.

1. **C1: force self-consistency (FSC)** as a label-free diagnostic, defined in §III and evaluated on every split.
2. **C2: a matched-budget audit** of three geometry-encoding families (graph, SDF-conditioned FNO, geometry transformer) under one protocol, one data pipeline and one training budget, including data-efficiency curves and targeted ablations.
3. **C3: calibration under shift**: deep ensembles plus split conformal prediction, with coverage, interval width and ECE reported per split and the exchangeability assumption stated where it breaks.
4. **C4: a solver-verified loop**, in which an acquisition score built from ensemble spread and FSC selects candidate cases that are then run in Ansys Fluent, a solver independent of the one that generated the training data, with a grid-independence study and a measured solver offset so the comparison is interpretable.

II. RELATED WORK

a. Neural operators and geometry encoding. Fourier neural operators [2], their geometry-informed extension [3], and the general operator-learning framework [4] together with DeepONet [5] and mesh-based graph networks [6] define the design space. Transolver [7] represents the attention-based branch.

b. Benchmarks for flow around complex geometry. AirfRANS [8] supplies the dataset, the metric definitions and the Reynolds and angle-of-attack splits this study inherits. Rabeh *et al.* [1] is the closest prior art and the framing anchor: it compares operators and foundation models across geometry encodings. Our axis is consistency and calibration rather than an accuracy ranking, so we cite it as motivation rather than as a competitor. DrivAerNet++ [9] provides the 3D extension. Takamoto *et al.* [10] and Ohana *et al.* [11] set the benchmark-documentation conventions we follow.

c. Physics-consistency losses. Physics-informed operators [12] penalise PDE residuals during training. We use the same machinery diagnostically: residuals are reported whether or not they are penalised, which is what separates a measurement from a training trick.

d. Uncertainty quantification. Deep ensembles [13] for the predictive spread; split conformal prediction [14–16] for distribution-free intervals, with Romano *et al.* [17] as an alternative score. Tibshirani *et al.* [18] is why our shape-family results are framed as a documented failure of exchangeability rather than as a bug. Vinuesa and Brunton [19] frames the CFD-community reception.

III. PROBLEM SETUP AND THE CONSISTENCY PROTOCOL

This section is the conceptual core of the paper: the shared operator-learning formulation, the force-integration route to a coefficient, the consistency diagnostic built from the gap between that route and a learned head, and the shift axes along which every model is evaluated.

A. Operator learning setup

Let $\Omega_g \subset \mathbb{R}^d$ be the fluid domain exterior to a body with boundary Γ_g , and let c be the flow condition; for the airfoil case $c = (U_\infty, \alpha)$ with $\text{Re} = U_\infty L / \nu$. We seek

$$\mathcal{S} : (g, c) \mapsto u = (\mathbf{v}, p, \nu_t)|_{\Omega_g}, \quad (1)$$

restricted in practice to the surface trace $\mathcal{S}_\Gamma : (g, c) \mapsto (c_p, \boldsymbol{\tau}_w)|_{\Gamma_g}$, since surface pressure and wall shear stress are what forces depend on.

B. Force integration

With $\mathbf{n}$ the outward normal and ΔS_i the facet measure taken from the fixed input geometry,

$$\mathbf{F} = \oint_{\Gamma_g} (-p \mathbf{n} + \boldsymbol{\tau}_w) \, dS \approx \sum_{i=1}^{N_f} \left(-p_i \mathbf{n}_i + \boldsymbol{\tau}_w^{(i)} \right) \Delta S_i, \quad (2)$$

$$C_D^{\text{int}} = \frac{\mathbf{F} \cdot \mathbf{e}_\infty}{\frac{1}{2} \rho U_\infty^2 A_{\text{ref}}}, \quad C_L^{\text{int}} = \frac{\mathbf{F} \cdot \mathbf{e}_\perp}{\frac{1}{2} \rho U_\infty^2 A_{\text{ref}}}. \quad (3)$$

Because ΔS_i and $\mathbf{n}_i$ come from the geometry and not from the prediction, (2) is *linear* in the predicted surface fields and differentiable, so it costs essentially nothing to evaluate or to backpropagate through.

Nothing in this paper is interpretable unless this integrator reproduces the dataset’s own reported coefficients on ground-truth fields, so we gate on it before any model result is reported: evaluated on all 1000 AirFRANS simulations, the median relative error is 3.3×10^{-4} on C_D (95th percentile 1.2×10^{-3} , maximum 1.6×10^{-3}) and 4.8×10^{-6} (median) on C_L : a factor of 30 inside the 1% tolerance we set for this gate, and about two orders of magnitude below the field errors reported in §VI. Every downstream FSC number therefore reflects the model, not the quadrature.

C. Force self-consistency

With C^{head} from a direct regression head on the same backbone,

$$\text{FSC} = |C^{\text{int}} - C^{\text{head}}|, \quad \text{FSC}_{\text{rel}} = \frac{|C^{\text{int}} - C^{\text{head}}|}{|C^{\text{true}}| + \varepsilon}. \quad (4)$$

FSC needs no ground truth. That is the whole point: it is computable on out-of-distribution splits and on unlabelled proposed geometries, which is what makes it usable inside the active-learning loop of §VII. When the force term is switched on during training (§IV, ablated rather than default), it penalises all three disagreements symmetrically, $\mathcal{L}_{\text{force}} = |C^{\text{int}} - C^{\text{true}}| + |C^{\text{head}} - C^{\text{true}}| + |C^{\text{int}} - C^{\text{head}}|$, so pulling either route toward the truth also pulls them toward each other.

D. Symmetry, no-slip and divergence residuals

With R the reflection about the relevant plane and $R^\sharp$ its action on vector fields, for $R\Gamma_g = \Gamma_g$ and a symmetric condition,

$$\mathcal{L}_{\text{sym}} = \|\hat{u}(g, c) - R^\sharp \hat{u}(Rg, Rc)\|_2 / \|\hat{u}(g, c)\|_2. \quad (5)$$

We also check antisymmetry on a symmetric section at $\pm\alpha$: $C_L(-\alpha) \approx -C_L(\alpha)$ and $C_D(-\alpha) \approx C_D(\alpha)$. Two further residuals are defined identically for every model and reported whether or not they are penalised. No-slip, with $\mathcal{N}_\delta$ the volume sample points within δ of the wall and $\omega(x) = 1 - \text{dist}(x, \Gamma_g)/\delta$ a linear falloff weight,

$$\mathcal{L}_{\text{bc}} = \frac{1}{|\mathcal{N}_\delta|} \sum_{x \in \mathcal{N}_\delta} \|\hat{\mathbf{v}}(x)\|^2 \omega(x). \quad (6)$$

Divergence, by least-squares gradients on the point cloud (or finite differences on M2-SDF-FNO’s latent grid),

$$\mathcal{L}_{\text{div}} = \frac{1}{|\mathcal{X}|} \sum_x (\nabla \cdot \hat{\mathbf{v}}(x))^2. \quad (7)$$

Every residual in this section ($\mathcal{L}_{\text{sym}}$, the antisymmetry gap, $\mathcal{L}_{\text{bc}}$ and $\mathcal{L}_{\text{div}}$) is computed and logged for every trained model regardless of whether its λ weight in §IV is zero, precisely so that a residual reported as a diagnostic is never confused with one that was optimised away.

One caveat on how the symmetry harness realises R matters for reading §VI A: the harness reflects *every* test geometry about the chord line and keeps per-point index correspondence. For a cambered airfoil the reflection is an inverted-camber shape that lies outside the NACA training family, so the residual on those samples mixes true equivariance violation with ordinary out-of-family extrapolation, and the large absolute magnitudes partly reflect the latter. The clean form of the test (zero-camber sections at $\alpha = 0$, where $Rg = g$) is the honest pure-equivariance number and is left as a follow-up; the residual as reported is best read as a response-to-reflection indicator whose *growth under shift* is the meaningful signal.

E. Shift magnitude

Each split’s test simulations are compared against *that split’s own* training support along the numeric design axes Re , α and section thickness. For axis j with training range $[\ell_j, h_j]$ and population standard deviation σ_j (over that split’s $\text{train} \cup \text{test}$),

$$\Delta_j(x) = \max\left(0, \frac{x - h_j}{\sigma_j}, \frac{\ell_j - x}{\sigma_j}\right), \quad (8)$$

zero for a test point inside the training box and otherwise its distance in units of that axis’s own spread, so the axes are commensurable. Shape family is categorical rather than metric (4-digit and 5-digit NACA sections are different parameterisations), so it contributes a binary term: $\Delta_{\text{shape}} = 1$ if the test simulation’s series is absent from training, else 0, one “unit” of shift on the same scale as one standard deviation of a numeric axis. The per-simulation shift magnitude is the Euclidean norm over these terms, $\Delta = (\Delta_{\text{Re}}^2 + \Delta_{\alpha}^2 + \Delta_{\text{thickness}}^2 + \Delta_{\text{shape}}^2)^{1/2}$, so a point in support on every axis scores zero and **combined** (shifted on two axes at once) lands furthest out by construction. Averaged over each split’s test set, the realised values are $\Delta \approx 0.00, 0.00, 0.42, 0.23, 1.00, 1.10$ for **full**, **scarce**, **reynolds**, **aoa**, **shape5** and **combined** (Table I). Note that this places **reynolds** numerically further from training support than **aoa**, and **shape5** furthest of all the single-axis shifts, yet §VIB finds the flow-regime shifts the more damaging ones: a first hint that raw design-space distance and model-relevant difficulty are not the same thing, since a shift in α or Re can cross into a qualitatively new flow regime (separation) that a Euclidean distance in design parameters cannot see. Figure 4 therefore plots error against split identity, ordered by shift axis, rather than against the continuous Δ value.

IV. MODELS AND TRAINING

We compare three geometry-encoding families under one matched protocol: the same shared coefficient head, the same loss family, the same optimiser and schedule, the same per-epoch checkpointing, and parameter counts held

within $\pm 20\%$ of a 1.5M target, so that architecture is the only thing that differs between runs. Realised counts are 1,389,637 (M1-GNN), 1,373,605 (M2-SDF-FNO) and 1,281,701 (M3-TRANSOLVER): an 8% max-to-min spread, well inside the budget.

a. M1-GNN: graph baseline. k -nearest-neighbour ($k = 16$) message passing on the surface point cloud with node features $(\mathbf{x}_i, \mathbf{n}_i, \kappa_i, c)$: position, outward normal, local (signed Menger) curvature, and the broadcast flow condition.

b. M2-SDF-FNO: SDF-conditioned FNO. A GINO-style kernel encoder scatters surface features onto a 64×64 regular latent grid conditioned on s_g , an FNO2d stack (16 Fourier modes, width 32) acts there, and a kernel decoder queries back to arbitrary surface points [2, 3].

c. M3-TRANSOLVER: geometry transformer. Transolver-style physics attention [7]: soft-assign N points to $M = 32$ slices across 4 attention layers at width 128, attend among slice tokens, scatter back, at cost $\mathcal{O}(NM + M^2)$.

d. Reimplementation. All three architectures are reimplemented from the cited papers rather than run from the authors' released code. The target hardware (a 4 GB Pascal-generation laptop GPU, with training moved to a cloud T4 once the study outgrew it) cannot reliably install the usual compiled geometry-learning dependencies (PyTorch Geometric, `torch_scatter`, `neuraloperator`) on this stack, so every graph, spectral-convolution and slice-attention operator here is written in plain PyTorch. A reviewer should not read any number in this paper as a reproduction of a published result: the comparison that is meaningful is the *within-study* one, matched by construction across the three families, not an absolute comparison against the architectures' original papers.

e. Training. Adam with a cosine learning-rate schedule, initial learning rate 10^{-3} , gradient-norm clipping at 1.0, batch size 16 for all three architectures (matched, not per-model tuned), 400 epochs, fp32 throughout (no mixed precision: the Pascal-generation hardware has no practical fp16 tensor-core path). Every run checkpoints every epoch and resumes from its own last checkpoint, which is what makes training across interruptible free-tier cloud sessions viable. Normalisation statistics (inputs and targets) are computed from *that split's own* training list only, never from a shared global set and never from calibration or test data, so no OOD split's test statistics can leak into training even indirectly through the input scaling.

f. Objective.

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{data}} + \lambda_H \mathcal{L}_{\text{head}} + \lambda_F \mathcal{L}_{\text{force}} \\ & + \lambda_B \mathcal{L}_{\text{bc}} + \lambda_S \mathcal{L}_{\text{sym}} + \lambda_D \mathcal{L}_{\text{div}}, \end{aligned} \quad (9)$$

each auxiliary term normalised by its value at initialisation. The coefficient head term is always on at $\lambda_H = 0.1$ (without it the head receives no gradient and FSC would be meaningless); the physics terms default to $\lambda_F = \lambda_B = \lambda_S = \lambda_D = 0$ and are ablation flags (§VIF), not part of the headline models.

V. DATASETS AND SPLITS

AirFRANS [8]: 1000 steady incompressible RANS solutions (Spalart–Allmaras) over NACA 4- and 5-digit sections, roughly 177,000 volume mesh points per case in the raw dataset (our cache subsamples the volume to at most 32,768 points) and $\mathcal{O}(10^3)$ surface points, $\text{Re} \in [2, 6] \times 10^6$, $\alpha \in [-5^\circ, +15^\circ]$, licensed **ODbL-1.0** (attribution and share-alike on derived databases; we release split manifests and code, never a processed copy of the fields). Every simulation encodes its own design point (inlet velocity, angle of attack, NACA parameters) in its filename, which is what makes the shift-magnitude computation of §III and the new splits below possible without any additional metadata.

Everything in this paper is AirFRANS, 2D and incompressible. DrivAerNet++ [9] (licensed CC BY-NC 4.0) is the natural 3D extension, but it requires an interactive data-access step outside this study's scope and contributes zero results here; no claim in this paper should be read as extending to 3D geometry, compressible flow, or any solver family other than the ones used (§VIII).

a. Splits. Six splits, all derived deterministically from the dataset's own manifest or from parsed design parameters, each with train/calibration/test lists whose pairwise disjointness is asserted by a dedicated test (`test_splits_no_leakage.py`): in-distribution **full** (the official random 800/200 train/test task); **scarce** (the official low-data task, 200 training cases, evaluated on **full**'s test set, which has no test set of its own); Reynolds shift **reynolds** (train $\text{Re} \in [3, 5] \times 10^6$, test outside that band); angle-of-attack shift **aoa** (train $\alpha \in [-2.5^\circ, 12.5^\circ]$, test outside, crossing toward separation at the highest angles); shape-family shift **shape5** (train every NACA 4-digit simulation, test every 5-digit one, a family split AirFRANS itself does not provide); and **combined** (train 4-digit *and* inside the Reynolds training band, test 5-digit *and* outside it: shape and Reynolds shifted together, the hardest cell, by design a strict strengthening of **reynolds**). Calibration sets are the last $\min(100, \max(1, \lceil 0.2 n_{\text{train}} \rceil))$ entries of each split's own seed-0-shuffled training list, so they scale with the split and never overlap train or test; test is never touched by normalisation or calibration (§IV).

TABLE I. Dataset and split card: per-split sizes for train, calibration and test, mean shift magnitude Δ over the test set (§III), source dataset, and licence.

Split	Train	Cal	Test	$\bar{\Delta}$	Source	Licence
full	700	100	200	0.00	AirFRANS	ODbL-1.0
scarce	160	40	200	0.00	AirFRANS	ODbL-1.0
reynolds	404	100	496	0.42	AirFRANS	ODbL-1.0
aoa	704	100	196	0.23	AirFRANS	ODbL-1.0
shape5	391	98	511	1.00	AirFRANS	ODbL-1.0
combined	191	48	246	1.10	AirFRANS	ODbL-1.0

TABLE II. In-distribution accuracy, consistency and cost (AirFRANS **full**). C_D MAE and FSC C_D in units of 10^{-3} . Neural rows are seed mean $\pm$ std over the 3–5 committed seeds; baselines are deterministic single fits. Bold marks the best neural model per column. The symmetry residual is Figure 6.

model	p rel- L_2	τ_w rel- L_2	C_D MAE	$C_D \rho$	FSC C_D	GPU-h
M1 GNN	0.0753 ± 0.0027	0.0686 ± 0.0021	0.132 ± 0.015	0.996 ± 0.0034	5.39 ± 0.088	2.88 ± 0.44
M2 SDF-FNO	0.0259 ± 0.0013	0.0307 ± 0.00088	0.142 ± 0.013	0.998 ± 0.0005	1.62 ± 0.2	0.36 ± 0.0022
M3 Transolver	0.0186 ± 0.00093	0.0192 ± 0.00081	0.135 ± 0.014	0.999 ± 0.00036	1.29 ± 0.31	0.408 ± 0.00076
Ridge	1.42	1.16	1.6	0.902	6.04	0
Constant	1.42	1.16	3.42	–	6.3	0

VI. RESULTS

Unless stated otherwise, every number below is a seed mean over the 3–5 committed seeds of that (model, split) cell, matching Tables II–III; **scarce** and **combined** are single-seed (no ensemble was trained on them).

A. In-distribution

On the in-distribution split **full**, all three architectures are accurate and the ordering is clean (Table II): field relative L_2 on surface pressure runs from 0.019 (M3-TRANSOLVER) through 0.026 (M2-SDF-FNO) to 0.075 (M1-GNN), a $4\times$ span that holds throughout the study. M3-TRANSOLVER $>$ M2-SDF-FNO $>$ M1-GNN on field accuracy, everywhere, matching the operator-learning literature’s expectation that attention over learned slices and spectral mixing on a regular grid both out-resolve local k NN message passing at matched budget. On C_D rank correlation from the coefficient head (the quantity a designer actually uses for down-selection) the picture inverts: all three sit at $\rho_s = 0.996$ – 0.999 on **full**, a spread of three parts per thousand. The **ridge baseline**, a linear regression from the shape and flow parameters directly to C_D with 29 parameters and zero training cost, is reported honestly rather than hidden: it reaches $\rho_s = 0.902$ in distribution, worse than every neural model but far from a naive floor, and (as §VIB shows) it holds up strikingly well under shift. All three architectures also show a large, roughly uniform symmetry residual (≈ 4.5 , Figure 6), consistent with the expectation that none of these encodings is architecturally equivariant to reflection; none uses the symmetry loss or augmentation during the runs reported here (§IV), so this is the raw, unmitigated residual, subject to the reflected-camber caveat of §III.

B. Out-of-distribution

Figure 4 and Table III give the full 3×6 grid (three models, six splits). The hierarchy is consistent and physical at the extremes, for every architecture: **full** is easiest and **combined** is hardest, and **shape5** and **scarce** sit as the two mildest single-axis shifts, in that order. The two remaining single-axis shifts, **reynolds** and **aoa**, are consistently the two *hardest* single-axis splits but do not settle into one fixed order between them: M2-SDF-FNO and M3-TRANSOLVER both find **aoa** harder than **reynolds**, while M1-GNN finds the reverse. We report this rather than round it into a clean ranking; it is a genuine architecture-dependent disagreement about which axis hurts more, sitting inside an otherwise architecture-independent hierarchy. M3-TRANSOLVER degrades from 0.019 (**full**) to 0.139 (**combined**), a $7.5\times$ increase; M1-GNN degrades from 0.075 to 0.270, only $3.6\times$: the weaker in-distribution model is, in relative terms, *more* robust to shift, so ranking by in-distribution accuracy alone would misorder the architectures for a deployment that includes shift.

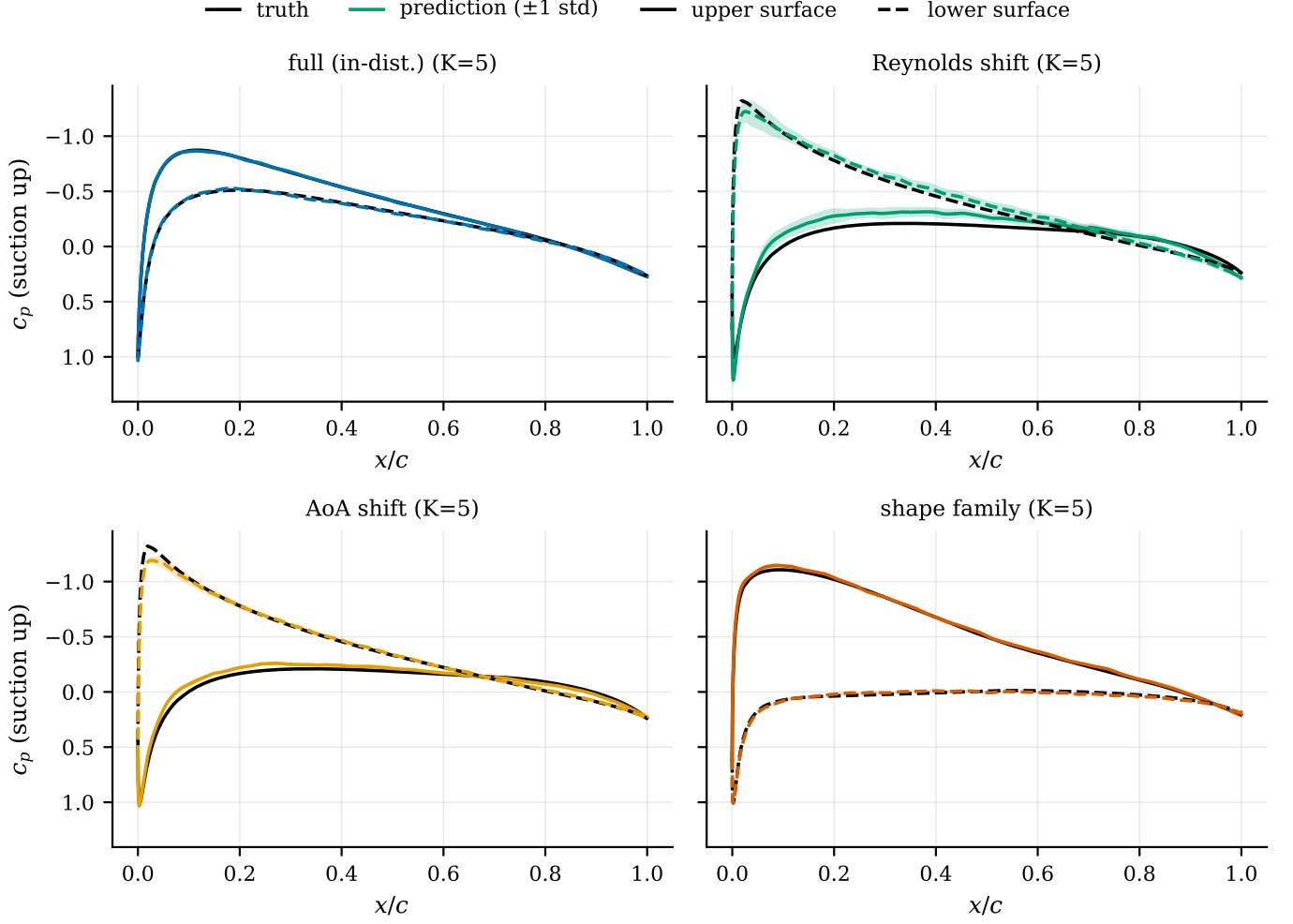


FIG. 2. M3-TRANSOLVER surface c_p , prediction versus truth, on one representative test simulation per split (full, Reynolds shift, AoA shift, shape family), upper and lower surface shown separately (solid/dashed), c_p axis inverted (suction up, the aerodynamics convention). The shaded band is the $K = 5$ deep-ensemble ± 1 standard deviation. In distribution the ensemble mean is visually indistinguishable from truth and the band is a hairline; under Reynolds and AoA shift the mean develops a systematic offset near the suction peak and the band widens but (exactly the pattern behind §VID’s coverage degradation) not enough to keep the truth inside it; under shape-family shift (a new NACA digit family, same flow regime) the fit stays close and the band stays narrow, consistent with **shape5** being the mildest single-axis shift for every architecture (§VIB).

Notably this field-error ordering does not track the shift-magnitude ordering of §III: **shape5** carries the single largest per-axis Δ in the whole study ($\Delta = 1.00$ by construction) while being the mildest single-axis shift for every model, and the flow-regime shifts ($\bar{\Delta} = 0.42$ and 0.23) are the two hardest. A purely geometric shift measure and the models’ actual difficulty ranking disagree, and the natural explanation is physical rather than statistical: α and Re shift move the flow toward conditions (higher effective angle of attack, different boundary-layer character) that can cross into a qualitatively new regime, whereas a new NACA digit family at the same Re and α is still an attached-flow problem the model has seen the physics of, just not the exact shape. These models interpolate within a flow regime and degrade gracefully doing so, but degrade sharply once the physics itself changes character. This reading rests on the aggregate split numbers plus the domain knowledge that the **aoa** test range crosses $\alpha \approx 12\text{--}15^\circ$, close to typical NACA-section stall onset at these Reynolds numbers; a per-simulation separation indicator would make it precise and is left as follow-up.

On head-route C_D rank correlation the between-model gap stays below 0.01 on five of the six splits and peaks at 0.03 on **reynolds** (where M3-TRANSOLVER’s seeds spread most), while the between-split swing for a single model reaches 0.05 ($0.999 \rightarrow 0.94$); shift moves the ranking metric more than architecture does, everywhere. The ridge baseline is, again, the uncomfortable honest number: $\rho_s \in [0.83, 0.90]$ across all six splits, meaning a 29-parameter linear model captures most of the head-route C_D -ranking signal that these ~ 1.3 M-parameter models deliver.

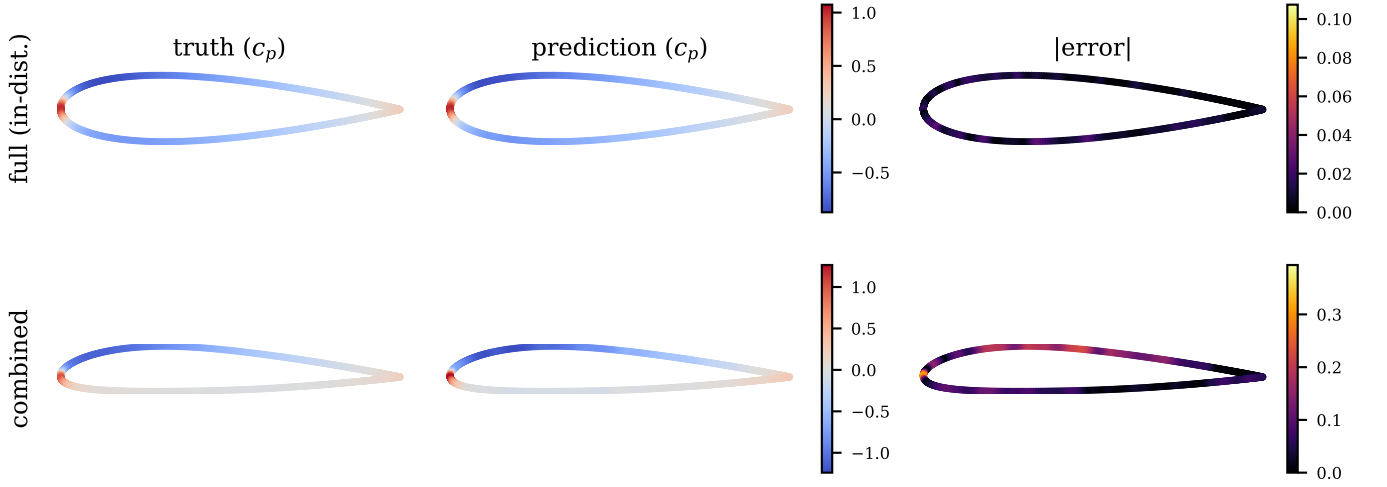


FIG. 3. M3-TRANSOLVER surface c_p scattered on the airfoil contour: truth, prediction, and absolute error (truth and prediction share one colour scale per row) for one in-distribution simulation (**full**) and one on the hardest shift (**combined**, shape and Reynolds shifted together), seed 0. The error is visibly concentrated near the leading edge, where the combined-shift suction peak is most mispredicted.

TABLE III. Out-of-distribution field p rel- L_2 by split (seed mean $\pm$ std where multiple seeds exist; **scarce/combined** are single-seed). Bold marks the best neural model per split.

model	full (in-dist.)	scarce	Reynolds shift	AoA shift	shape family	combined
M1 GNN	0.0753 ± 0.0027	0.154	0.214 ± 0.012	0.195 ± 0.0055	0.128 ± 0.0036	0.27
M2 SDF-FNO	0.0259 ± 0.0013	0.0832	0.104 ± 0.015	0.113 ± 0.0058	0.0665 ± 0.0026	0.153
M3 Transolver	0.0186 ± 0.00093	0.0666	0.0837 ± 0.0063	0.101 ± 0.007	0.0526 ± 0.0025	0.139
Ridge	1.42	1.4	1.62	1.18	1.6	1.86
Constant	1.42	1.4	1.62	1.18	1.6	1.86

The **integrated** drag ranking is the sharper counterpoint. Ranking test cases by C_D^{int} (drag obtained by integrating the *predicted* field, seed means as in the tables) gives $\rho_s = 0.89$ (M3-TRANSOLVER), 0.83 (M2-SDF-FNO) and 0.36 (M1-GNN) on **full**, and collapses to 0.28, -0.02 and -0.28 respectively on **combined**. So the field-consistent route ranks drag *no better than ridge* for the best architecture in distribution and far worse for the graph net, and fails entirely under the combined shift. Any “beats ridge” statement in this paper is therefore a head-route statement; the drag implied by the predicted physics does not beat ridge. This is Gap 1 landing hard: the models learn a good drag regressor and a good field, but not a field whose integral *is* that drag, which is exactly what §VIC quantifies.

C. Consistency

The absolute FSC numbers in Table II look small only because C_D itself is ~ 0.01 . As a fraction of the true drag (median FSC_{rel}), the coefficient head and the integrated field disagree by $\approx 5\%$ (M3-TRANSOLVER), 7% (M2-SDF-FNO) and 24% (M1-GNN) already in distribution, rising to $\approx 37\%$, 51% and 115% on **combined**. The two “drags” a single model reports are tens of percent apart, and further apart the more it extrapolates: that is why the integrated ranking in §VIB is so much weaker than the head ranking.

Pooled across all 58 committed core cells (three models, six splits, all seeds), FSC_{rel} on C_D correlates strongly with field relative L_2 (Pearson $r = 0.89$, Spearman $\rho = 0.87$): both quantities respond to the same underlying cause, distribution shift. That pooled correlation, however, is mostly a statement about shift, not about the models: holding the split fixed, M3-TRANSOLVER is both the most accurate and the most self-consistent model on **full**, **scarce**, **shape5** and **combined**. On **aoa**, however, M2-SDF-FNO is the more self-consistent model (seed-mean FSC 0.0034 vs. 0.0040) even though M3-TRANSOLVER remains the more accurate one in field norm; on **reynolds** the two are statistically tied at seed means (0.0056 vs. 0.0055; seed 0 alone puts M2-SDF-FNO ahead there, but the seed means do not support it). That is a concrete instance of the hypothesis this metric is meant to test: the field-error leaderboard and the consistency leaderboard disagree, and they disagree on a flow-regime shift, exactly where a designer relying on the field-best model’s reported drag would care most. FSC_{rel} also grows from **full** to **combined** at least as fast

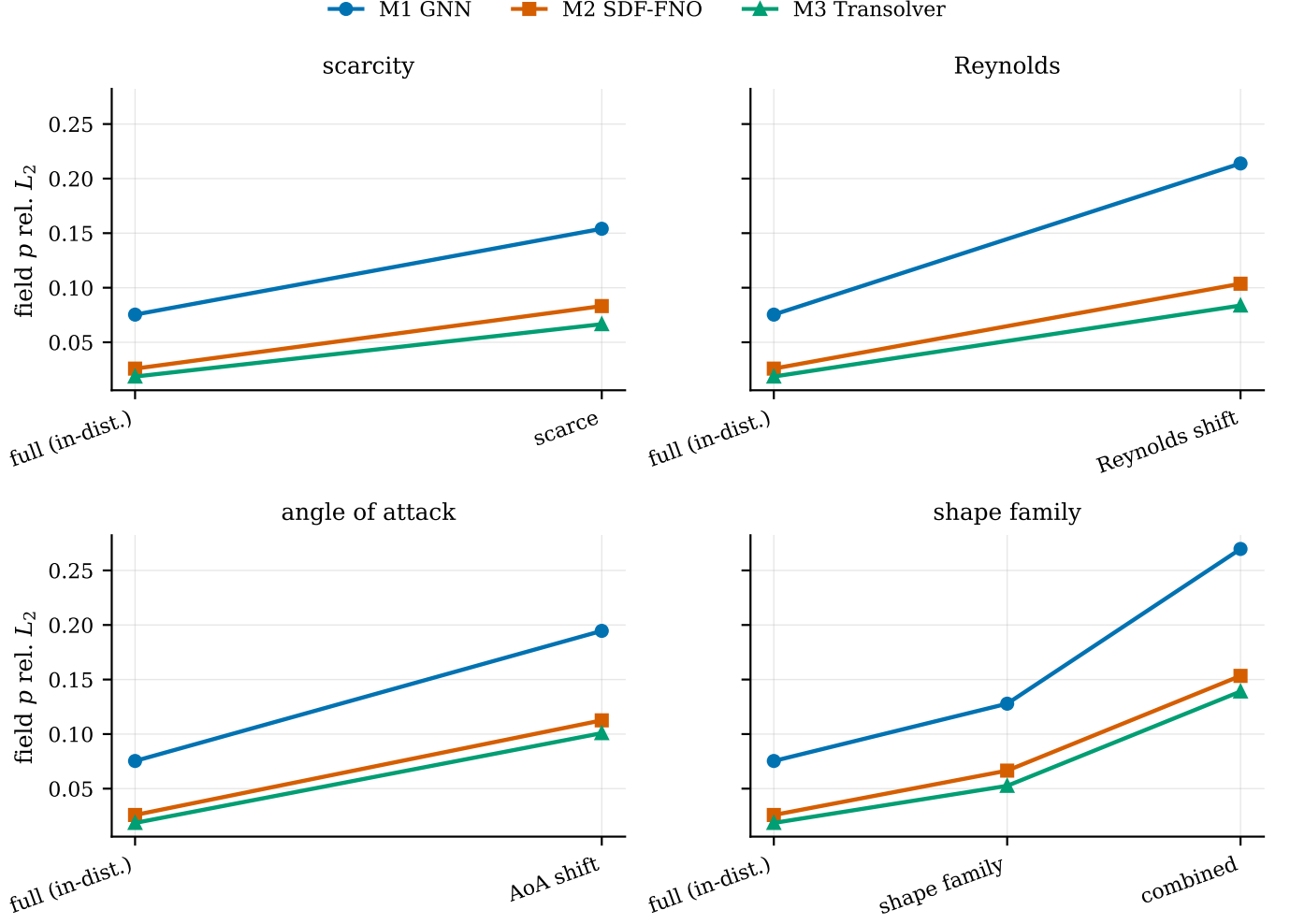


FIG. 4. Error versus distribution shift, one panel per OOD axis (scarcity, Reynolds, angle of attack, shape family/combined), one line per model, splits ordered along each axis; seed means over the committed core runs. Field relative L_2 on surface pressure.

as the field error it accompanies ($4.9\times$ vs. $3.6\times$ for M1-GNN, $6.9\times$ vs. $5.9\times$ for M2-SDF-FNO, $8.0\times$ vs. $7.5\times$ for M3-TRANSOLVER), so self-consistency is an equally or more sensitive shift indicator than the field metric, while needing no ground truth: exactly what makes it usable as a label-free OOD flag and as an input to the acquisition score of §VII.

D. Calibration

Table IV and Figures 7–9 report split-conformal calibration of the integrated drag band C_D^{int} at nominal 80/90/95%, matched (calibrate and test on the same split’s own sets) and transfer (calibrate on `full`, deploy on the shifted split). Ensembles cover $\{\text{full}, \text{reynolds}, \text{aoa}, \text{shape5}\}$: $K = 5$ for M2-SDF-FNO and M3-TRANSOLVER, $K = 3$ for M1-GNN (the k NN graph net costs 7–8 $\times$ more per run, so it gets a smaller ensemble by design). One wording caveat up front: each split’s calibration set is carved from that split’s own *training* pool, so “matched” means a like-distributed calibration set is available, the optimistic case; it is not recalibration on labelled cases from the shifted deployment itself, which is not run here.

On a calibration set of n simulations disjoint from train and test, with normalised nonconformity score $s_j = |y_j - \bar{u}(\mathbf{x}_j)| / (\hat{\sigma}(\mathbf{x}_j) + \varepsilon)$ and $\hat{q}$ its $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile,

$$\mathcal{C}_\alpha(\mathbf{x}) = [\bar{u}(\mathbf{x}) - \hat{q} \hat{\sigma}(\mathbf{x}), \bar{u}(\mathbf{x}) + \hat{q} \hat{\sigma}(\mathbf{x})], \quad (10)$$

which satisfies $\mathbb{P}(y \in \mathcal{C}_\alpha(\mathbf{x})) \geq 1 - \alpha$ when calibration and test data are exchangeable. An *absolute* score ($s_j =$

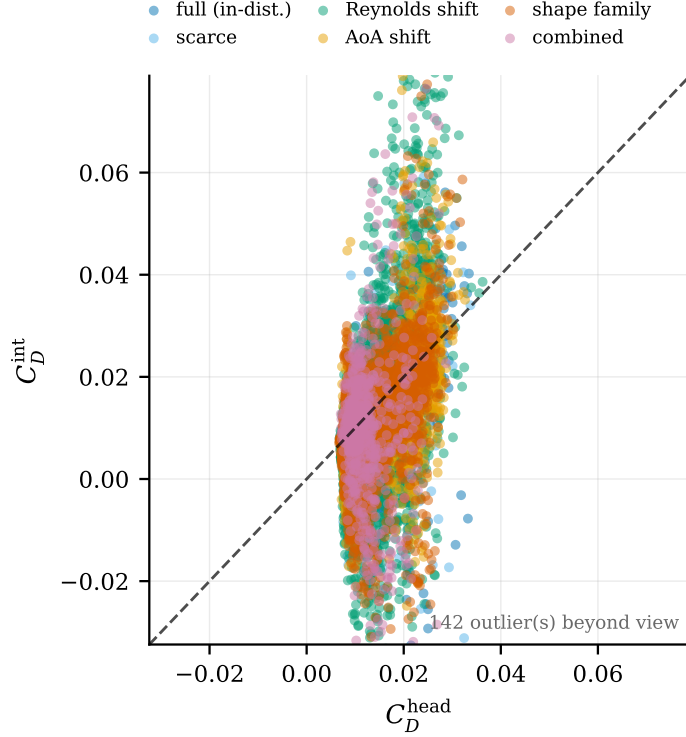


FIG. 5. Force self-consistency: C_D^{int} against C_D^{head} , one point per test simulation pooled over all three models, coloured by split, against the identity line (axes zoomed to the central 99% of the data). Distance from the identity line is FSC; the vertical spread at fixed C_D^{head} is the integrated route disagreeing with the head it shares a backbone with.

$|y_j - \bar{u}(\mathbf{x}_j)|$, a fixed-width band) is calibrated in parallel, and the contrast between the two matters below. Two granularities are calibrated separately: per-simulation coefficient intervals, where the exchangeable unit is a whole simulation and the guarantee is clean; and pointwise field intervals, where points within a simulation are *not* exchangeable, so we conformalise at the simulation level with a quantile-over-points score and state the weaker interpretation rather than implying a pointwise guarantee.

In distribution, matched calibration is close to nominal for the two stronger models but not automatic: coverage at nominal 90% is 0.955 (M2-SDF-FNO) and 0.91 (M3-TRANSOLVER), while M1-GNN *under-covers* at 0.82 (Wilson 95% interval $[0.76, 0.87]$, excluding 0.90). “The conformal guarantee holds in distribution” is therefore true for two of three models here, not all three.

Under shift, a high coverage number can be an artifact of band width, and the honest reading requires both columns of Table IV. The clearest case is M1-GNN on **reynolds**: matched coverage 0.97 looks like a calibration win, but the normalized band behind it has mean width 0.228 for a C_D of ~ 0.01 , an interval $\approx 23\times$ the quantity it brackets: near-vacuous. The tight absolute-score band on the same split covers only 0.70, essentially unchanged from a single member, so the ensemble did not genuinely fix the graph net under shift; it inflated the variance-scaled interval. We report both scores rather than quote the flattering one. The stronger models fail elsewhere: M2-SDF-FNO under-covers on **shape5** (0.763 at nominal 90, its worst split), the shift §VIB found mildest for field accuracy but evidently not for calibration; M3-TRANSOLVER is the most consistently calibrated of the three (matched coverage 0.87–0.95 everywhere) but also dips on **shape5** (0.867).

ECE degrades most on the deployment-relevant absolute-score bands. On the absolute score, ECE jumps 6–10 $\times$ from **full** to the flow-regime shifts (M1-GNN $0.022 \rightarrow 0.16$, M2-SDF-FNO $0.047 \rightarrow 0.27$, M3-TRANSOLVER $0.040 \rightarrow 0.44$). Table IV’s ECE column reports the normalized-score ECE, which moves far less (e.g. M3-TRANSOLVER $0.007 \rightarrow 0.035$) precisely because the normalized band widens with the ensemble variance: the same band-width effect as the coverage row, stated here so the table cannot be over-read.

Matched versus transfer calibration does not order cleanly. Naively, matched calibration (the split’s own, in-principle-exchangeable calibration set) should be the safer choice. The data does not cooperate: on **shape5**, transfer beats matched for M2-SDF-FNO (0.847 vs. 0.763) but loses to it for M1-GNN (0.818 vs. 0.863) and M3-TRANSOLVER (0.814 vs. 0.867); on **reynolds** the normalized matched numbers exceed transfer for M1-GNN and M3-TRANSOLVER while M2-SDF-FNO orders the other way (0.895 vs. 0.940). No single choice of calibration set is

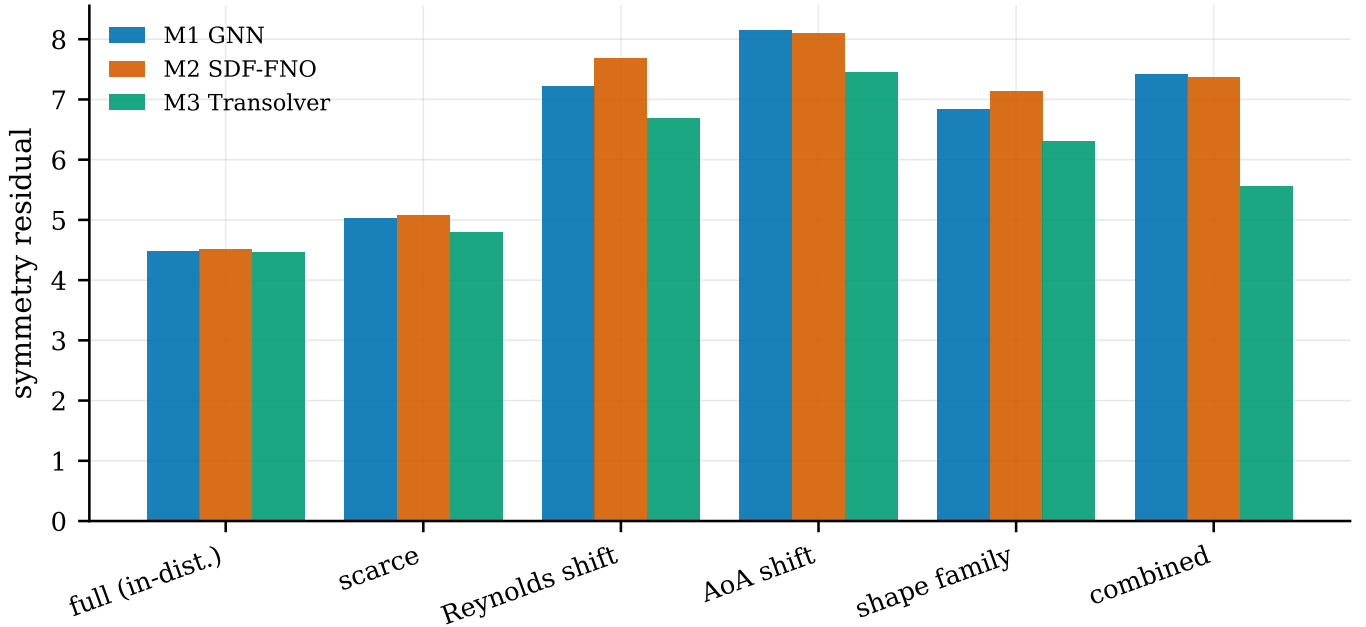


FIG. 6. Symmetry (mirror-equivariance) residual per model, by split; no augmentation is applied in any run reported here (§IV). The residual grows from **full** to **aoa** for every architecture and never approaches zero, confirming that none of the three encodings is built to be equivariant to reflection; M3-TRANSOLVER is consistently the least violating, M1-GNN and M2-SDF-FNO track each other closely. Read magnitudes with the reflected-camber caveat of §III; the growth under shift is the robust signal.

reliably safer once exchangeability has broken: evidence that the failure is structural (the score’s relationship to error has genuinely changed under shift) rather than a fixable artifact of which calibration set was used.

Interval width widens under shift for every model (Figure 9), but under-compensates. C_D^{int} band width at nominal 90% grows from **full** to the harder splits (e.g. M2-SDF-FNO $0.0085 \rightarrow 0.022$, M3-TRANSOLVER $0.0056 \rightarrow 0.021$ on **reynolds**, roughly $2.5\text{--}3.7\times$). That the uncertainty estimate responds to shift at all is informative (an unresponsive score would be worse), but coverage still drops on the same splits where width grows most, which is what “widening honestly but insufficiently” looks like in a reliability diagram (Figure 7).

The conclusion, stated as plainly as the numbers allow: split conformal prediction repairs in-distribution calibration for a well-behaved model with a like-distributed calibration set, but its coverage guarantee is not robust to distribution shift. A 90% conformal interval computed on this study’s splits must not be read as a 90% probability statement once the input has moved outside the calibration distribution: under a flow-regime shift the deployment-relevant absolute-score coverage can be as low as 0.70 in this study, and a high normalized-score coverage may be reporting nothing more than a very wide band. Practically, a conformal interval should be trusted only for inputs a label-free monitor (such as FSC, §VIC) has flagged as in-distribution, and a reported coverage should always be read next to its interval width.

E. Data efficiency and cost

All three models were retrained at training-set sizes $\{25, 50, 100, 200, 400\}$, three seeds each (45 runs), with the full-700 point reused from the core grid: Figure 10(a). Two things are worth the reader’s attention. First, **M3-Transolver is the most accurate at every size**, including 25 airfoils (field p rel- L_2 0.271 vs. 0.292 for M1-GNN and 0.325 for M2-SDF-FNO): the transformer’s advantage is not a large-data artifact. Second, **the M2-SDF-FNO-vs-M1-GNN order crosses**: the graph net is the better of the two in the scarce regime ($n = 25$: 0.292 vs. 0.325; $n = 50$: 0.217 vs. 0.232) and M2-SDF-FNO overtakes it only around $n \approx 70\text{--}100$, after which its spectral route scales much faster (at $n = 700$: 0.026 vs. 0.075). The crossing is clear on the seed means; in seed-noise terms the $n = 50$ gap is $\approx 1.6\text{--}2.4$ standard deviations over three seeds while the $n = 25$ gap is within one, so the statement “M2-SDF-FNO is the most data-hungry of the three below ~ 100 airfoils” is solid from $n = 50$ up and indicative at $n = 25$. The clean M3-TRANSOLVER $>$ M2-SDF-FNO $>$ M1-GNN hierarchy of §VIA is therefore a large-data statement, and a single-budget ranking under-specifies method choice for data-scarce practice.

TABLE IV. Split-conformal calibration of the integrated drag band (C_D^{int} , normalized score, largest available ensemble: $K=3$ for the GNN, $K=5$ otherwise): empirical coverage, mean interval width and ECE. Matched calibrates on the split’s own cal set; transfer (tr) calibrates on full’s cal. Coverage must be read next to width: the GNN’s high shifted coverage rides on a near-vacuous band (§VID).

model	split	cov@.8	cov@.9	cov@.95	cov@.9 (tr)	width@.9	ECE
M1 GNN	full (in-dist.)	0.735	0.82	0.92	0.82	0.0332	0.0583
	shape family	0.718	0.863	0.959	0.818	0.0467	0.0426
	Reynolds shift	0.933	0.966	0.972	0.917	0.228	0.0737
	AoA shift	0.816	0.893	0.929	0.893	0.0575	0.015
M2 SDF-FNO	full (in-dist.)	0.885	0.955	0.98	0.955	0.00854	0.0567
	shape family	0.648	0.763	0.879	0.847	0.00731	0.12
	Reynolds shift	0.75	0.895	0.938	0.94	0.0218	0.0224
	AoA shift	0.913	0.944	0.974	0.929	0.0159	0.0605
M3 Transolver	full (in-dist.)	0.805	0.91	0.955	0.91	0.00562	0.00667
	shape family	0.793	0.867	0.908	0.814	0.00746	0.0275
	Reynolds shift	0.76	0.938	0.978	0.915	0.021	0.0351
	AoA shift	0.821	0.949	0.959	0.949	0.0198	0.0265

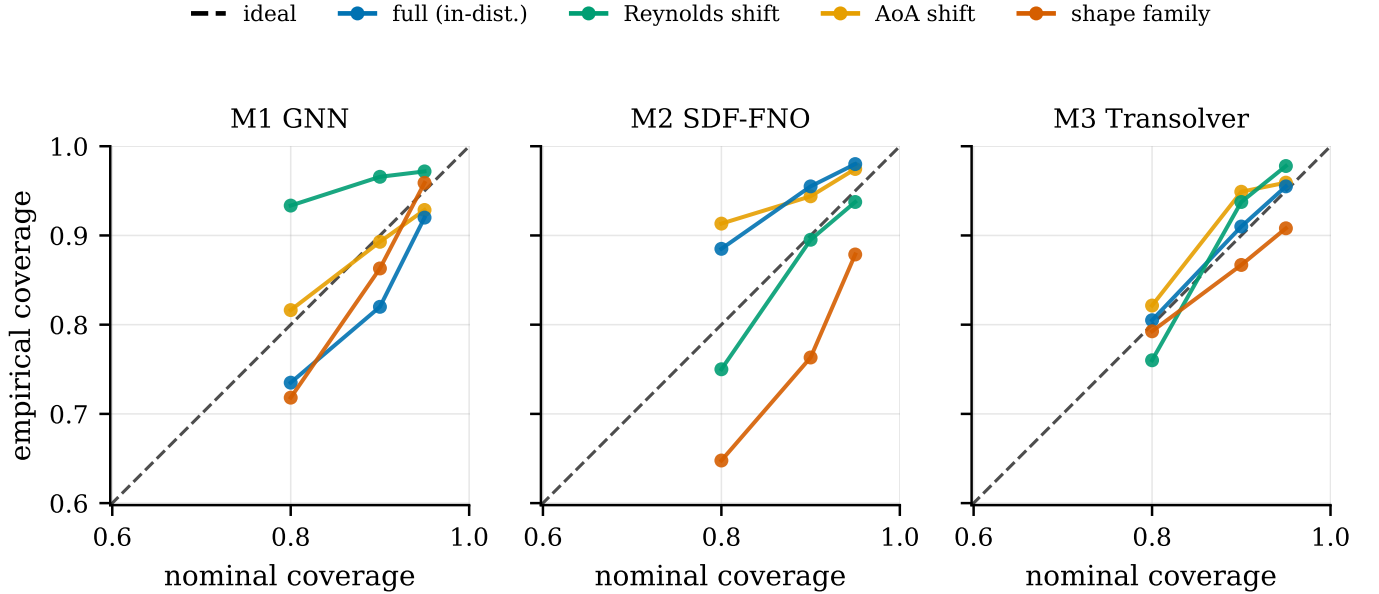


FIG. 7. Reliability of the matched-calibration C_D^{int} band, one panel per model, coloured by split, largest available K per model ($K = 3$ M1-GNN, $K = 5$ otherwise): empirical versus nominal coverage at 80/90/95%. Curves below the diagonal under-cover.

The cost axis (Figure 10(b)) makes the deployment reading direct: at the full budget M1-GNN is Pareto-dominated, costing 2.4–3.2 GPU-h across seeds (mean 2.9) to train against M2-SDF-FNO’s 0.36 h and M3-TRANSOLVER’s 0.41 h (the k NN reconstruction on every forward pass dominates its cost) while also being the least accurate. The efficient frontier is M3-TRANSOLVER and M2-SDF-FNO; M1-GNN’s appeal below $n \approx 100$ (above) is the one place this study found it earning its cost.

F. Ablations

Table V reports the two ablation axes run for this version (seed 0). **Geometry conditioning for M2-SDF-FNO:** the binary occupancy mask is the worst encoding on both splits (0.029 in distribution, 0.073 on *shape5*), SDF is best in distribution (0.026), and SDF plus surface normals gives the best out-of-family accuracy (0.061 vs. 0.064 for SDF alone). This reproduces the published finding that SDF beats a mask [1] and adds that normals help specifically under shape-family shift. **Force-consistency loss for M1-GNN:** switching λ_F on cuts FSC by 4 $\times$ in distribution (0.0053 $\rightarrow$ 0.0013) and 3 $\times$ on *combined*, while *worsening* in-distribution field accuracy (0.074 $\rightarrow$ 0.104): the physics

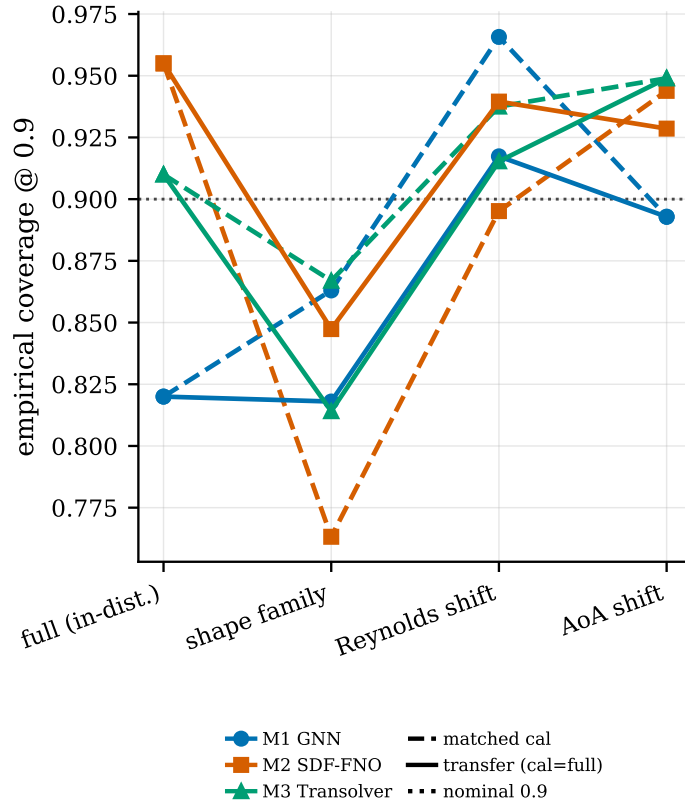


FIG. 8. Empirical coverage at nominal 90% against split, matched (dashed) versus transfer-calibrated on **full** (solid), normalized score. Neither calibration choice reliably dominates once the split shifts, and M1-GNN’s high **reynolds** point rides on a near-vacuous band width (§VID).

TABLE V. Ablations (seed 0). (a) M2-SDF-FNO geometry conditioning: field p rel- L_2 in distribution and under the shape-family shift. (b) M1-GNN force-consistency loss (λ_F): the physics term cuts the residual it penalises (FSC) at a cost in in-distribution field accuracy.

(a) M2-SDF-FNO geometry conditioning (p rel- L_2)				
conditioning	full	shape5 (OOD)		
SDF (baseline)	0.0257	0.06414		
binary mask	0.02935	0.07279		
SDF + normals	0.03092	0.06127		

(b) M1-GNN force-consistency loss				
variant	full p rel- L_2	full FSC	combined p rel- L_2	combined FSC
baseline ($\lambda_F = 0$)	0.07374	0.0053	0.2697	0.01761
+ force loss ($\lambda_F > 0$)	0.1043	0.001333	0.259	0.005926

term shrinks exactly the residual it penalises and buys self-consistency, not accuracy. This is the predicted negative result, reported as found. The ensemble-size axis is implicit in the $K \in \{1, 3, 5\}$ conformal reports of §VID (the M1-GNN $K = 1 \rightarrow 3$ comparison there is its most instructive data point); the augmentation ablation for the symmetry residual was cut from this version’s scope.

VII. ACTIVE LEARNING WITH SOLVER VERIFICATION

This section turns FSC and ensemble spread from diagnostics into a use: a label-free score that picks which of many un-simulated candidates is worth the cost of a new CFD run, verified against a solver (Ansys Fluent 2021 R1) that did not generate the training data. The outcome is stated honestly up front: the score concentrated its picks on the

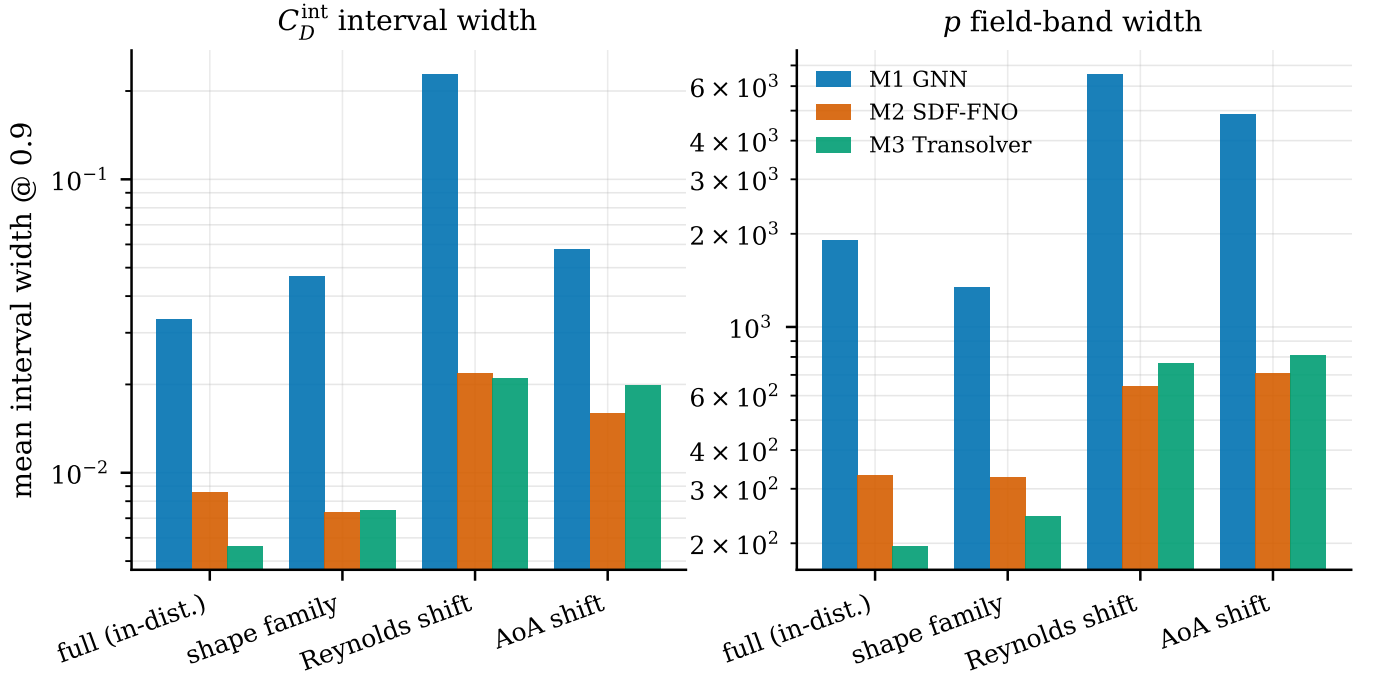


FIG. 9. Interval-width distributions, in-distribution versus OOD, matched calibration: mean interval width at nominal 90%, per model and split, for the C_D^{int} coefficient band and the pointwise field band (log scale; the field band is in kinematic-pressure units). Widths grow from full outward for every model, but, as Table IV and Figure 8 show together, not by enough to keep coverage at nominal under shift.

corner of the pool where the steady operator the surrogate learned mostly has no settled solution, so the quantitative comparison is a qualitative external check on the solvable subset, not a large-sample statistical test.

A. Acquisition

The pool is a parametric family of NACA 4- and 5-digit sections crossed with a grid of (Re, α) : 630 candidates, $\text{Re} \in \{2, \dots, 7\} \times 10^6$, $\alpha \in \{-6, 0, 6, 12, 18\}^\circ$, none coinciding with a training case. Each candidate is scored

$$a(g, c) = \underbrace{\hat{\sigma}_{C_D}(g, c)}_{\text{spread}} + \beta \underbrace{\text{FSC}(g, c)}_{\text{self-inc.}} + \gamma \underbrace{\mathcal{L}_{\text{sym}}(g, c)}_{\text{sym. viol.}}, \quad (11)$$

all three normalised to unit variance over the pool, with $\beta = \gamma = 1$. The top- k are selected subject to greedy farthest-point diversity in normalised design space. Three arms of $k = 8$: acquisition, pure ensemble variance, and random. Scorer: the M3-TRANSOLVER $K = 5$ full ensemble.

Figure 11(a) shows the result before any solver runs: the acquisition and variance arms place **all 16 of their picks at** $\alpha = 18^\circ$, the top of the pool grid and well past both the training range (12.5°) and the dataset's own envelope ($\sim 15^\circ$), while the random arm spreads across the envelope. Two disclosures qualify this. First, the score is built from the *integrated* C_D on the synthetic pool contour, and that representation carries a large artifact off the real mesh (pool $\sigma_{C_D} \approx 0.15\text{--}0.24$ and $\text{FSC} \approx 0.08\text{--}0.31$ for a C_D of ~ 0.03 , versus ~ 0.001 on real meshes in distribution; the root cause was not isolated and is logged as open). Second, the pool grid overshoots the training envelope on both axes by construction, so the extreme corner was nearly guaranteed to top a ranking that rewards extrapolation. The defensible claim is therefore that the label-free score **flags the out-of-envelope corner**: it detects extrapolation, which here coincides with hard physics, and the solver runs below test what that is worth.

B. Solver setup, grid independence, and the solver offset

Every selected case runs as 2D steady incompressible RANS on a parameterised structured C-grid, wall-resolved ($y_{\text{max}}^+ \leq 1.1$ on every accepted case), with Spalart–Allmaras [20] as the like-for-like closure (matching AirFRANS's

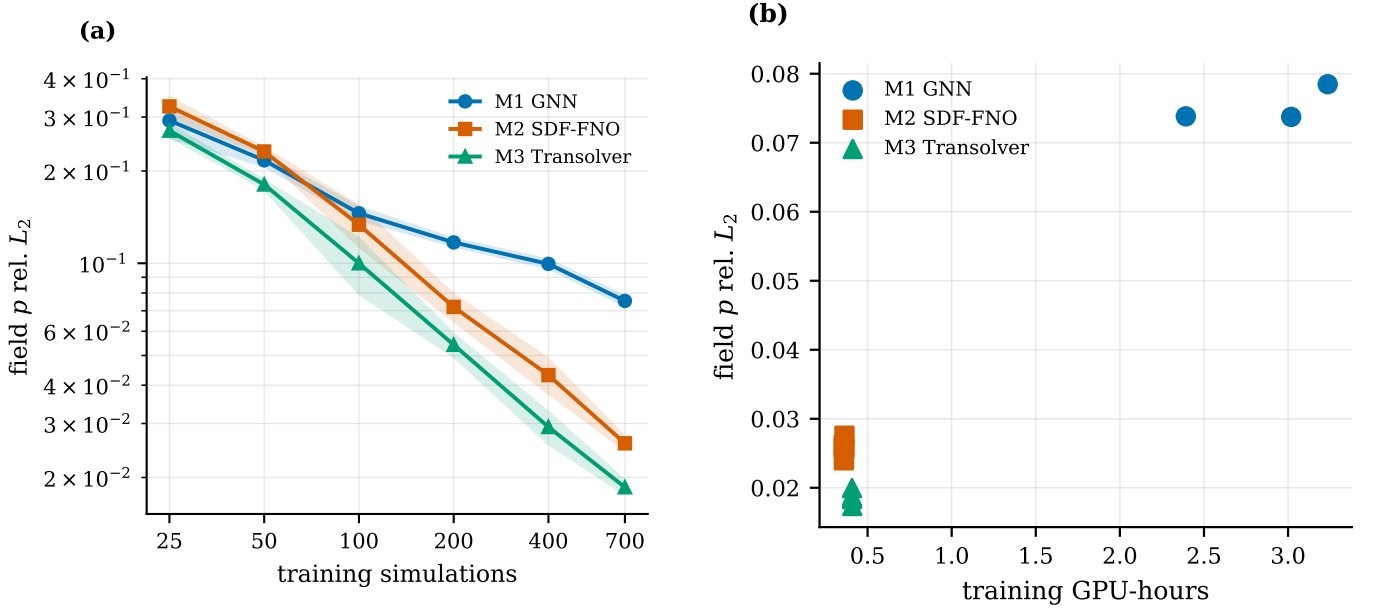


FIG. 10. Data efficiency and the cost-accuracy trade-off. (a) Data efficiency: field error against training-set size, log-log, seed bands shaded. The M2-SDF-FNO/M1-GNN curves cross around $n \approx 70$ –100 training simulations; M3-TRANSOLVER leads everywhere. (b) Cost versus accuracy at the full training budget.

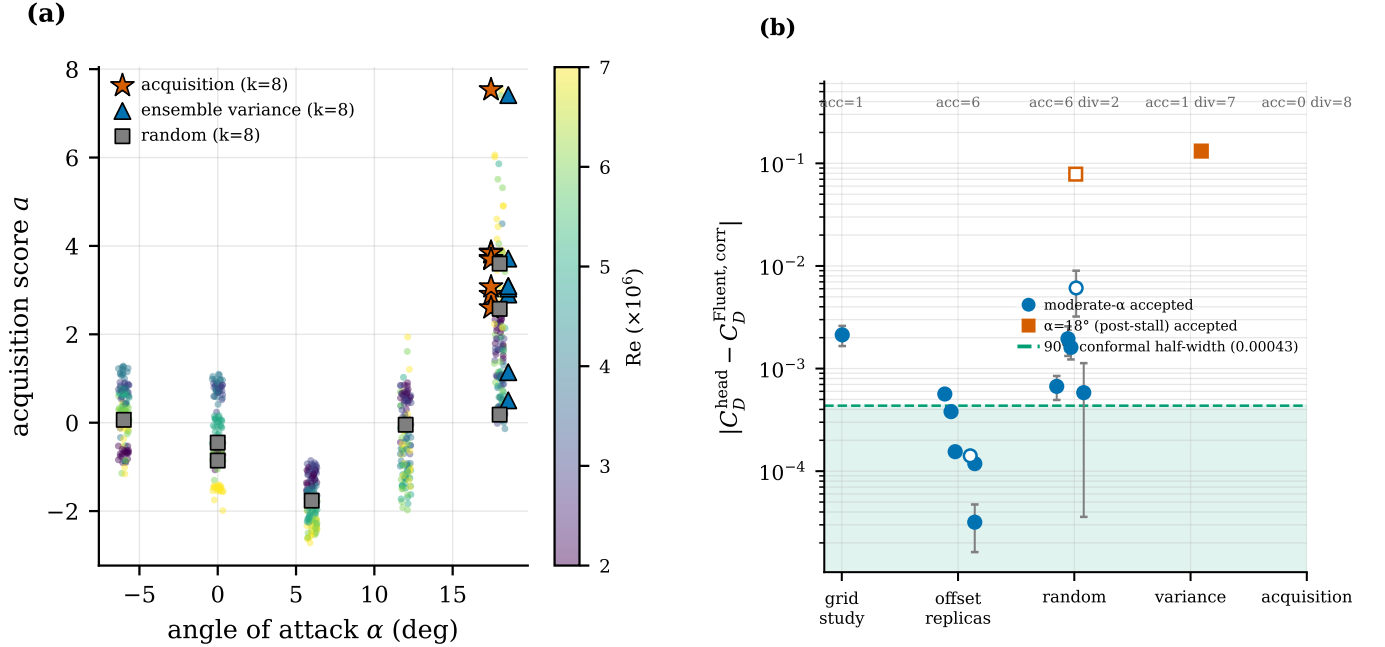


FIG. 11. Active learning: (a) the 630-candidate pool against α , coloured by Re , with the three $k=8$ arms marked; acquisition and pure-variance selection agree with each other (all 16 picks at $\alpha = 18^\circ$) and disagree with random. (b) M3-TRANSOLVER $|C_D|$ error against the offset-corrected Fluent value on the accepted set, by arm and regime; the annotations count accepted (acc) and non-settling (div) cases per arm, and the shaded band is the 90% conformal half-width from the full calibration study.

own generation) and k - ω SST [21] run per case for a model-sensitivity spread. A three-level grid study on NACA0012 at $Re = 3 \times 10^6$, $\alpha = 5^\circ$ (18,796 / 43,008 / 96,768 cells) gives C_D changing 3.3% from L1 to L2 and 0.003% from L2 to L3 with C_L within 0.3% and $y^+ < 1$ everywhere: grid-independent at L2, the level every production case uses (Appendix B reports the Celik GCI [22, 23] and the sanity checks). The SA-SST C_D spread at L2 is 1.5%.

TABLE VI. Solver offset: six AirfRANS test simulations re-solved in Fluent (SA, grid level L2). $\delta C_D = C_D^{\text{Fluent}} - C_D^{\text{AirfRANS}}$; every replica reads high, so the mean offset $\bar{\delta}_{C_D}$ is subtracted before any surrogate comparison.

replica (NACA)	C_D^{AF}	C_L^{AF}	C_D^{SA}	C_D^{SST}	δC_D	rel. δC_D
4 (2710)	0.009226	0.0513	0.009719	0.009553	0.000493	0.0534
1 (0016)	0.009702	0.251	0.01048	0.01019	0.00078	0.0804
2 (1009)	0.01151	0.926	0.01301	0.01304	0.0015	0.13
6 (5_17)	0.009805	0.219	0.01082	0.01032	0.00101	0.103
3 (2211)	0.009052	0.607	0.009919	0.009786	0.000867	0.0958
5 (0010)	0.008058	0.419	0.00881	0.008599	0.000752	0.0933
mean $\pm$ s (SA, n=6):						+0.0009 $\pm$ 0.00034

Because Fluent and the OpenFOAM setup behind AirfRANS differ in solver, discretisation and mesh topology, six AirfRANS *test* simulations were replicated in Fluent (geometry gate: maximum nearest-neighbour distance to the cached surface below 8×10^{-5} chords) to measure the solver offset before interpreting any surrogate-versus-Fluent gap (Table VI). On SA, $\bar{\delta}_{C_D} = +0.00090$ ($s = 0.00034$, $n = 6$; every replica positive, 5.3–13% of C_D), and $\bar{\delta}_{C_L} = +0.0122$. Crucially, the per-case SA-SST spread (0.00003–0.0005) is *smaller* than the offset, so at these angles the solver difference, not the turbulence model, is the dominant systematic; the mean SA offset is subtracted from every Fluent C_D before comparison. The replica scatter $s = 0.00034$ is itself as large as the surrogates’ in-distribution C_D MAE (~ 0.00012 – 0.00014), a solver-side floor that the comparison below cannot go beneath.

C. Campaign outcome

Of the 33 cases run (3 grid levels, 6 replicas, 24 arm picks), the grid trio, all 6 replicas, 4 low- α random picks and (on retry) the $\alpha = 12^\circ$ random pick converged or reached a clean quasi-steady state. **All 20 first-attempt divergences sat at the aggressive corner:** the 16 acquisition- and variance-arm picks (all $\alpha = 18^\circ$), 3 random $\alpha = 18^\circ$ picks, and one random $\alpha = 12^\circ$ case. A conservative retry (longer first-order start, reduced under-relaxation, no early stopping) split that set three ways: the in-envelope $\alpha = 12^\circ$ case recovered cleanly ($C_D = 0.019$), confirming a settings failure rather than physics; **two $\alpha = 18^\circ$ cases genuinely settled** (the variance-arm pick, converged $C_D = 0.167$ with drift 0.2%, and a random-arm case, quasi-steady $C_D = 0.113$); and of the remaining 17, sixteen ran their full 7000 iterations at bounded, physically plausible post-stall magnitudes (final $C_D \approx 0.06$ – 0.15) while still drifting 7–21% over the last 500 iterations and one errored out at iteration 725, so all 17 are classified non-settling: past the envelope where this steady-RANS operator has a solution to converge to. A time-averaged URANS truth for those 17 is future work, and would compare the surrogate to a quantity it was never trained to predict.

D. Surrogate versus Fluent

For every accepted case the trained ensembles were evaluated at that geometry and condition and compared to the offset-corrected Fluent coefficient (Table VII; per-case rows in Appendix Table X). The reported surrogate C_D is the coefficient head; the integrated route on the synthetic pool contour carries the representation artifact disclosed above, so it is kept only as a diagnostic. The accepted set has two regimes, and averaging them into one MAE would hide the entire finding.

Moderate α (n=12, -6° to 12°): surrogate C_D error is 3–10 $\times$ its in-distribution level (MAE 0.00037 for M1-GNN, 0.0012 M3-TRANSOLVER, 0.0013 M2-SDF-FNO against in-distribution ~ 0.00012 – 0.00014), and the 90% conformal intervals under-cover for two of three models (M3-TRANSOLVER 0.42, M2-SDF-FNO 0.33; M1-GNN’s 0.83 rides on its wider band, §VID). Verified against a solver that did not generate the training data, the C3 calibration degradation reproduces against external truth. Part of this gap is the solver-side floor above, which is negligible against what follows.

Post-stall (n=2, the settled $\alpha = 18^\circ$ cases): the surrogates predict attached-flow drag ($C^{\text{head}} \approx 0.03$ – 0.06) where the offset-corrected settled Fluent values are 0.113 and 0.165: errors of 0.05–0.13 per case, roughly 640–900 $\times$ the in-distribution MAE, every one far outside its conformal band.

TABLE VII. Surrogate vs. Fluent on the accepted steady set, split by regime (all cases at the grid-independent L2 level, Appendix B; qualitative external check, §VII). Surrogate C_D is the coefficient head, compared against the offset-corrected Fluent value; cov@90 is the fraction inside that model’s 90% conformal interval from the full-split calibration study. In-dist. MAE is the same model’s C_D MAE on the full test set.

model	moderate α ($-6^\circ..12^\circ$)				post-stall ($\alpha = 18^\circ$)	
	n	MAE C_D	ratio to in-dist.	cov@90	n	MAE C_D
M1 GNN	12	0.00037	$3\times$	0.83	2	0.1
M2 SDF-FNO	12	0.0013	$9.2\times$	0.33	2	0.087
M3 Transolver	12	0.0012	$10\times$	0.42	2	0.11

E. The C4 finding

The falsifiable claim was: cases with high acquisition score have higher surrogate error against independent truth than randomly selected ones. The verdict splits cleanly by half. The half that says *high-score picks are cases the surrogate fails on* is now supported with a hard number: the one high-score ($\alpha = 18^\circ$ variance-arm) pick the solver settled carries a surrogate error orders of magnitude above the moderate-regime error level, and the other 15 high-score picks sit where the steady solver itself does not settle, the strongest possible form of “the surrogate should not be trusted here.” (The second settled $\alpha = 18^\circ$ case belongs to the random arm and fails identically, which cuts against the other half below.) The half that says *random picks are cases the surrogate gets right* is only partly true: on the tightly controlled offset replicas the surrogate sits inside its band in 15 of 18 model-case pairs, but on the random arm coverage is 3/6 (M1-GNN) and 0/6 (M2-SDF-FNO, M3-TRANSOLVER). So the score does separate hard from easy, the conformal bands do not survive the trip, and both statements are visible at once in Figure 11(b).

VIII. DISCUSSION AND LIMITATIONS

Overclaiming is the single biggest risk to a paper whose subject is trust, so this section is written to be read, not skimmed.

a. Limitations.

- **Seed counts vary by cell.** Core-grid cells on `full`, `reynolds`, `aoa` and `shape5` carry 3–5 seeds (Tables II–III report seed mean $\pm$ std); `scarce` and `combined` are single-seed, so per-seed spread there is unknown. Ensembles are $K = 5$ for M2-SDF-FNO/M3-TRANSOLVER and $K = 3$ for M1-GNN (a cost decision, stated in §VID); no ensemble exists on `scarce` or `combined`, so no conformal result is claimed there.
- **One dataset family, 2D only.** Everything here is AirfRANS: 2D airfoils, incompressible RANS, Spalart–Allmaras upstream. No claim extends to 3D geometry, compressible flow, or other solvers. DrivAerNet++ remains future work (§V).
- **The “matched” calibration is the optimistic case.** Each split’s calibration set is carved from its own training pool (§VID); recalibrating on labelled cases from the shifted deployment itself is the practically interesting variant and is not run here.
- **The acquisition outcome is confounded by its own construction.** The score uses the integrated C_D on a synthetic pool contour that carries a large representation artifact (§VII; root cause open), and the pool grid overshoots the training envelope on both axes, so “all 16 picks at $\alpha = 18^\circ$ ” should be read as the score flagging the out-of-envelope corner, not as evidence that a calibrated field-level uncertainty located hard physics on its own.
- **The Fluent leg is an external check, not a statistical test.** Fourteen accepted cases, two of them post-stall; the solver offset is bounded by six replicas whose scatter ($s_{C_D} = 0.00034$) is as large as the surrogates’ in-distribution MAE and is not folded into the coverage test. Ten pool cases (two accepted) at $\text{Re} = 7 \times 10^6$ run at $M \approx 0.32$, past the usual incompressible limit; solver and surrogate are both incompressible, so the comparison is internally consistent, but absolute coefficients there carry that caveat.
- **The symmetry residual is a response-to-reflection indicator.** For cambered sections the reflected input is out-of-family (§III), so absolute magnitudes overstate pure equivariance violation; the zero-camber clean test is future work. (The cached signed curvature fed to the reflected batch was verified numerically identical to recomputing it on the mirrored contour, so the harness itself is sound.)

- **Reimplementations, not the authors’ code** (§IV): absolute numbers are not comparable with published ones; only the within-study comparison is meaningful.
- **Compute budget.** Free-tier cloud T4 sessions and a 4 GB laptop GPU, 61 GPU-hours total (Appendix C); every model trains exactly 400 epochs at batch 16. Rankings at a substantially larger budget may differ, which is exactly what the data-efficiency curves are for.
- **Single author.** No co-author caught bugs. Mitigations: the force integrator is gated against ground truth before anything downstream runs (§III); split disjointness is asserted programmatically for every manifest; and multiple independent automated review passes audited the codebase and the results, catching real defects that were fixed before this text was written (a normalisation leak across splits, a table aggregation that mixed ablation runs into core cells, and a divergence classifier that misread the iteration-1 startup transient; all disclosed in the repository’s decision log).

b. Negative and mixed results, reported as found. The ridge baseline (29 parameters, zero training cost) is competitive on head-route C_D rank correlation across every split ($\rho_s \in [0.83, 0.90]$), and the integrated drag ranking of the best neural model does not beat it at all (§VIB). The architecture that wins on field accuracy is not always the most internally self-consistent: M2-SDF-FNO out-consistents M3-TRANSOLVER on the AoA shift and ties it on the Reynolds shift (§VIC). The force-consistency training term buys self-consistency, not accuracy (§VIF). Matched-versus-transfer conformal calibration does not order cleanly (§VID). The M1-GNN ensemble upgrade ($K = 1 \rightarrow 3$) raised its normalized shifted coverage without moving the absolute-score coverage: a band-width effect, not a calibration gain, disclosed rather than quoted. And the acquisition arm’s falsifiable claim survived on one half and only partly on the other (§VII). Each of these would have been easy to leave out; a trust protocol that hides its own mixed results would be self-refuting.

IX. CONCLUSION

This paper’s contribution is a measurement protocol, not a leaderboard position. On raw field accuracy the three audited architectures separate cleanly (M3-TRANSOLVER > M2-SDF-FNO > M1-GNN, everywhere, by up to $4\times$), but on head-route C_D ranking they are within a few parts per thousand of each other in distribution (within a few hundredths under shift) and only modestly ahead of a 29-parameter ridge baseline, and on the physics-consistent integrated ranking the best of them does not beat that baseline at all. A field-error leaderboard, run in isolation, would have reported a result that barely survives contact with the metric a designer would use. That is the first transferable finding: *for down-selection tasks, measure the ranking metric, and measure it through the physics, before declaring an architecture the winner.*

The second finding is that self-consistency and calibration are not by-products of accuracy. The two drags a single model reports disagree by 5–24% in distribution and up to 115% under shift; the field-accuracy leader is not always the consistency leader (it loses that title on the AoA shift); and conformal coverage that looks healthy can be a wide band reporting nothing (§VID). Each has to be measured separately, and each disagrees with the accuracy ranking often enough to matter.

The third finding is about shift itself: every axis this protocol measures (field error, ranking, FSC, symmetry residual, coverage) degrades most on the two shifts that cross toward a new flow regime and only mildly on the shape-family shift, even though the latter carries the largest raw design-space distance. These models interpolate within a flow regime and do not extrapolate across one, a distinction design-space distance cannot see but the label-free FSC and ensemble spread track well enough to be used: in this study’s pool they concentrated the acquisition picks exactly where the independent solver later found either no steady solution or surrogate errors three orders of magnitude above nominal.

What a practitioner should take to work: report the ranking metric beside the field norm, and report it through the integrated physics, not only a supervised head. Compute FSC on every prediction, including in production on unlabelled shapes: it is free and it tracks shift about as well as the field error it proxies. Treat a conformal interval as calibrated only for inputs a label-free monitor confirms in-distribution, and never quote a coverage without its width. And when deciding where the next expensive simulation goes, an ensemble-and-consistency score is a cheap, effective extrapolation detector, provided its inputs are computed on a representation whose artifacts are understood.

REPRODUCIBILITY STATEMENT

Code is released under the MIT licence at https://github.com/cocomo069/Geometry_Aware_Neural_Operator_Audit. The release includes: split-ID manifests for all six splits (data/splits/*.json, the exact train/calibration/test

TABLE VIII. Architecture hyperparameters, one column per model.

	M1-GNN	M2-SDF-FNO	M3-TRANSOLVER
encoder width / channel dim	128 (hidden)	32 (FNO width)	128 (d)
depth	8 (msg. rounds)	4 (FNO layers)	4 (attn. layers)
geometry parameter	$k = 16$ (kNN)	16 modes, 64×64 grid	$M = 32$ slices
conditioning	pos./normal/curvature	signed distance (s_g)	8 heads, slice attn.
grouped/spectral detail	—	4 spectral groups; kernel enc/dec $k=8$	—
positional Fourier frequencies	6	6	6
coefficient head	trunk 128–256, 2-layer MLP, deep-set mean+max pooling (identical)		
parameter count	1,389,637	1,373,605	1,281,701

TABLE IX. Training hyperparameters, matched across all three architectures (§IV) and identical in every core-grid, ensemble, data-efficiency and baseline-comparable run.

Optimiser	Adam, weight decay 0
Learning rate	10^{-3} initial, cosine schedule to 10^{-6} , 0 warmup epochs
Gradient clipping	global norm 1.0
Batch size	16 (all three architectures)
Epochs	400
Precision	fp32 (no AMP; the Pascal-generation local GPU has no practical fp16 path)
Loss weights (default)	$\lambda_H = 0.1$ (coefficient head, always on); $\lambda_F = \lambda_B = \lambda_S = \lambda_D = 0$
Seeds	core grid 0–4 (or 0–2 for M1-GNN); data-efficiency 0–2; ensembles as in §VID
Normalisation	per-split, train-list-only statistics (§IV)
Checkpointing	every epoch, resumable from the last checkpoint

lists behind every result); per-run configs and metrics under `results/<run_id>/` (`config.yaml`, `metrics.json`, `per_sim.csv`) for every core, ensemble, data-efficiency, ablation and baseline run; the conformal calibration reports (`results/uq/*.json`) behind Table IV and Figures 7–9; the active-learning pool, scores and selection (`results/active/*`); and the Fluent verification set (meshes, journals, convergence histories and extracted results under `fluent/` and `results/fluent/`, CC BY 4.0). Every figure and table in this paper is produced by the committed scripts (`make_figures.py`, `render_examples.py` and `make_tables.py`, under `scripts/`) from the committed result files, not hand-typed, so a reader can regenerate every number from the release.

Data: AirfRANS [8] is licensed ODbL-1.0 (attribution and share-alike on derived databases); this release publishes split manifests and code, never a processed copy of the AirfRANS fields. DrivAerNet++ [9] contributes zero results here and is not included.

Normalisation (§IV): every split’s statistics come from that split’s own training list only. This is a deliberate fix to an earlier version of the pipeline that shared one global normalisation across splits, described honestly because it changed reported OOD numbers; the affected runs were retrained under the fix before any result in this paper was produced.

Seeds: every run records its seed in `metrics.json`; ensemble members are seeds 0–4 (M2-SDF-FNO/M3-TRANSOLVER) or 0–2 (M1-GNN). Training is deterministic given seed, split manifest and code commit, modulo GPU reduction-order non-associativity, which is not controlled across the local Quadro P2000 and the cloud Tesla T4.

Appendix A: Hyperparameters and training details

Full configs are committed under `configs/` (`gnn.yaml`, `sdf_fno.yaml`, `transolver.yaml`) and, per run, at `results/<run_id>/config.yaml`; this appendix reproduces the values needed to rebuild a run without opening those files.

Appendix B: CFD setup and grid-independence study

Written for a CFD reviewer, who reads this appendix first. The full plan, templates and per-case journals are in the repository under `fluent/`.

a. Meshing. `fluent/mesh_gen.py` writes the C-grid mesh directly as a native Fluent `.msh` from the NACA parametric equations, rather than through a general-purpose mesher (gmsh cannot emit Fluent’s format; `meshio`’s Fluent writer drops boundary zones): the topology is a single structured block with a known index map, so the y^+

TABLE X. Per-case surrogate vs. Fluent comparison on the accepted steady set; all drag values in units of 10^{-2} . $C_D^{\text{fl,corr}}$ is the offset-corrected Fluent drag; model columns are ensemble mean $\pm$ std of the coefficient head; cov@90 flags whether M3-TRANSOLVER lands inside its 90% conformal interval.

NACA	arm	Re	α	status	$C_D^{\text{fl,corr}}$	M3-TRANSOLVER	M2-SDF-FNO	M1-GNN	cov@90
0010	random	7	6	converged	0.889	1.05 ± 0.04	0.986 ± 0.03	0.906 ± 0.01	no
22012	random	3	12	quasi-steady	1.85	2.46 ± 0.3	2.1 ± 0.1	1.72 ± 0.04	no
2415	random	6	0	converged	0.856	0.923 ± 0.02	1.05 ± 0.03	0.879 ± 0.003	no
33012	random	6	-6	converged	0.93	0.988 ± 0.05	1.22 ± 0.4	1 ± 0.1	no
4412	random	7	18	quasi-steady	11.3	3.49 ± 0.2	5.93 ± 0.9	4.1 ± 0.9	no
4415	random	2	0	converged	1.08	1.27 ± 0.06	1.34 ± 0.08	1.1 ± 0.02	no
32012	variance	6	18	converged	16.5	3.37 ± 0.2	4.46 ± 0.6	3.77 ± 0.6	no
0012	gridstudy	3	5	converged	0.985	1.2 ± 0.05	1.25 ± 0.04	1.01 ± 0.01	no
0016	offset	3.91	2.4	converged	0.958	0.97 ± 0.0004	0.97 ± 0.001	0.97 ± 0.001	yes
1009	offset	4.04	8.58	converged	1.21	1.15 ± 0.005	1.15 ± 0.01	1.15 ± 0.0009	no
2211	offset	4.6	3.33	converged	0.902	0.905 ± 0.002	0.908 ± 0.003	0.905 ± 0.002	yes
2710	offset	2.06	-1.58	converged	0.882	0.92 ± 0.002	0.923 ± 0.007	0.925 ± 0.009	yes
0010	offset	6.01	3.79	converged	0.791	0.806 ± 0.001	0.806 ± 0.003	0.813 ± 0.002	yes
5-17	offset	4.14	0.932	quasi-steady	0.992	0.977 ± 0.0009	0.976 ± 0.002	0.981 ± 0.003	yes

target and the grid-refinement ratio are exact one-line parameters instead of sizing-function requests. `mesh_gen.py --self-check` re-parses the written mesh and asserts index ranges, owner/neighbour consistency and non-degenerate faces.

b. Solver setup. Ansys Fluent 2021 R1, 2D steady incompressible RANS, SIMPLE, second-order upwind after a first-order start; two closures per case (SA matching AirfRANS’s generation, $k-\omega$ SST for spread). Per case, the molecular viscosity is set so the Fluent freestream reproduces the target Re exactly. Convergence is judged on a windowed C_D history (peak-to-peak and drift over the last 500 iterations), never tuned per case; a case that does not meet the criteria is reported as non-settling, not massaged until it converges. One collector-level correction is disclosed: the blow-up test ($|C_D| > 10$) originally examined every iteration including the impulsive start (iteration 1 spikes to $|C_D| \approx 10$ –14 on every case before the field develops), which mis-flagged two genuinely settled $\alpha = 18^\circ$ retries; the test now skips the first 50 iterations, and that correction is what admitted the two post-stall points of §VIID.

c. Grid-independence study. NACA0012, $\text{Re} = 3 \times 10^6$, $\alpha = 5^\circ$, three levels:

level	cells	y_{max}^+	C_D (SA)	C_L (SA)
L1 (coarse)	18,796	0.87	0.011122	0.5538
L2 (medium)	43,008	0.59	0.010755	0.5530
L3 (fine)	96,768	0.39	0.010755	0.5521

C_D moves 3.3% from L1 to L2 and 0.003% from L2 to L3; C_L varies under 0.3%; $y^+ < 1$ at every level. The L2-L3 C_D difference sits at the round-off floor, so the formal Celik $\text{GCI}_{\text{fine}}(C_D, \text{SA}) \approx 0\%$ with a noise-dominated observed order; the honest numerical band is the L1-to-L2 change, 3.3%. SST is better behaved (GCI_{fine} : C_D 0.26%, C_L 0.22%; SA C_L 3.4%). Sanity: $C_L = 0.553$ against the thin-airfoil $2\pi\alpha = 0.548$ (within 1%), C_D in the expected 0.008–0.012 band. All production cases run at L2.

d. Solver offset. Table VI in the main text; the geometry gate and per-case viscosity matching are described in §VIIB.

Appendix C: Compute accounting

Every number below is summed directly from the `train_time_s` field of the 110 committed per-run `metrics.json` files that carry it (core grid, deep ensembles, data-efficiency sweep and ablations; the two non-learned baselines fit in sub-second CPU time and are excluded), so it is a lower bound on total compute, not an estimate:

Model	GPU-hours	runs summed
M1-GNN	42.8	32
M2-SDF-FNO	9.8	41
M3-TRANSOLVER	8.4	37
Total	61.0	110

TABLE XI. Active-learning acquisition-arm selection ($k = 8$ of 630 pool candidates, scorer `transolver_full_k5`, $\beta = \gamma = 1$): NACA section, Reynolds number, angle of attack, ensemble spread $\hat{\sigma}_{C_D}$, force self-consistency FSC, symmetry residual $\mathcal{L}_{\text{sym}}$, and the resulting acquisition score. Generated from `results/active/selection_summary.json` and `results/active/acquisition_ranking.csv`. Every pick sits at $\alpha = 18^\circ$; per §VII C, none of the eight settled on retry (seven ran their full 7000 iterations bounded but non-settling, one errored at iteration 725); the two settled $\alpha = 18^\circ$ cases belong to the variance and random arms. Score components are computed on the synthetic pool contour and carry the representation caveat of §VII.

NACA	Re	α ($^\circ$)	$\hat{\sigma}_{C_D}$	FSC	$\mathcal{L}_{\text{sym}}$	score	a
0010	7×10^6	18	0.171	0.312	0.745	7.533	
2410	5×10^6	18	0.178	0.153	0.794	3.858	
4410	7×10^6	18	0.159	0.168	0.759	3.840	
0010	3×10^6	18	0.195	0.124	0.933	3.714	
33012	7×10^6	18	0.171	0.154	0.747	3.692	
4415	7×10^6	18	0.185	0.122	0.719	3.072	
0015	6×10^6	18	0.240	0.077	0.751	2.901	
24015	7×10^6	18	0.187	0.100	0.733	2.597	

M1-GNN dominates the bill despite the fewest runs: k NN graph reconstruction on every forward pass makes it 7–8 $\times$ costlier per run (~ 2.9 h against 0.36/0.41 h for M2-SDF-FNO/M3-TRANSOLVER on `full`), which is also why it carries $K = 3$ ensembles rather than $K = 5$. Hardware: local gating and development on a Quadro P2000 (4 GB, Pascal); every training run summed above executed on Kaggle’s free-tier Tesla T4 (16 GB), chained across interruptible sessions via per-epoch checkpointing (§IV). The local machine’s CPU ran the Fluent leg (6 cores). Energy accounting is not available from the Kaggle API and is not reported.

REFERENCES

-
- [1] A. Rabeh, E. Herron, A. Balu, S. Sarkar, C. Hegde, A. Krishnamurthy, and B. Ganapathysubramanian, Benchmarking scientific machine-learning approaches for flow prediction around complex geometries, *Communications Engineering* **4**, 10.1038/s44172-025-00513-3 (2025).
 - [2] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar, Fourier neural operator for parametric partial differential equations, in *International Conference on Learning Representations (ICLR)* (2021) [arXiv:2010.08895 \[cs.LG\]](#).
 - [3] Z. Li, N. Kovachki, C. Choy, B. Li, J. Kossaifi, S. Otta, M. A. Nabian, M. Stadler, C. Hundt, K. Azizzadenesheli, and A. Anandkumar, Geometry-informed neural operator for large-scale 3D PDEs, in *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)* (2023) [arXiv:2309.00583 \[cs.LG\]](#).
 - [4] N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar, Neural operator: Learning maps between function spaces with applications to PDEs, *Journal of Machine Learning Research* **24**, 1 (2023), [arXiv:2108.08481](#).
 - [5] L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis, Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators, *Nature Machine Intelligence* **3**, 218 (2021), [arXiv:1910.03193](#).
 - [6] T. Pfaff, M. Fortunato, A. Sanchez-Gonzalez, and P. W. Battaglia, Learning mesh-based simulation with graph networks, in *International Conference on Learning Representations (ICLR)* (2021) [arXiv:2010.03409 \[cs.LG\]](#).
 - [7] H. Wu, H. Luo, H. Wang, J. Wang, and M. Long, Transolver: A fast transformer solver for PDEs on general geometries, in *International Conference on Machine Learning (ICML)* (2024) [arXiv:2402.02366 \[cs.LG\]](#).
 - [8] F. Bonnet, J. A. Mazari, P. Cinnella, and P. Gallinari, AirFRANS: High fidelity computational fluid dynamics dataset for approximating Reynolds-averaged Navier–Stokes solutions, in *Advances in Neural Information Processing Systems 35 (NeurIPS 2022) Datasets and Benchmarks Track* (2022) dataset licence ODbL-1.0, [arXiv:2212.07564 \[cs.LG\]](#).
 - [9] M. Elrefaie, F. Morar, A. Dai, and F. Ahmed, DrivAerNet++: A large-scale multimodal car dataset with computational fluid dynamics simulations and deep learning benchmarks, in *Advances in Neural Information Processing Systems 37 (NeurIPS 2024) Datasets and Benchmarks Track* (2024) dataset licence CC BY-NC 4.0, [arXiv:2406.09624 \[cs.LG\]](#).
 - [10] M. Takamoto, T. Praditia, R. Leiteritz, D. MacKinlay, F. Alesiani, D. Pflüger, and M. Niepert, PDEBench: An extensive benchmark for scientific machine learning, in *Advances in Neural Information Processing Systems 35 (NeurIPS 2022) Datasets and Benchmarks Track* (2022) [arXiv:2210.07182 \[cs.LG\]](#).
 - [11] R. Ohana, M. McCabe, L. Meyer, R. Morel, F. J. Agocs, M. Beneitez, M. Berger, B. Burkhart, S. B. Dalziel, D. B.

- Fielding, *et al.*, The well: a large-scale collection of diverse physics simulations for machine learning, in *Advances in Neural Information Processing Systems 37 (NeurIPS 2024) Datasets and Benchmarks Track* (2024) arXiv:2412.00568 [cs.LG].
- [12] Z. Li, H. Zheng, N. Kovachki, D. Jin, H. Chen, B. Liu, K. Azizzadenesheli, and A. Anandkumar, Physics-informed neural operator for learning partial differential equations, *ACM/JMS Journal of Data Science* (2024), arXiv:2111.03794 [cs.LG].
 - [13] B. Lakshminarayanan, A. Pritzel, and C. Blundell, Simple and scalable predictive uncertainty estimation using deep ensembles, in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)* (2017) arXiv:1612.01474 [stat.ML].
 - [14] A. N. Angelopoulos and S. Bates, A gentle introduction to conformal prediction and distribution-free uncertainty quantification, *arXiv preprint arXiv:2107.07511* (2021), arXiv:2107.07511 [cs.LG].
 - [15] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World* (Springer, 2005).
 - [16] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, Distribution-free predictive inference for regression, *Journal of the American Statistical Association* **113**, 1094 (2018).
 - [17] Y. Romano, E. Patterson, and E. J. Candès, Conformalized quantile regression, in *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)* (2019) arXiv:1905.03222 [stat.ME].
 - [18] R. J. Tibshirani, R. F. Barber, E. J. Candès, and A. Ramdas, Conformal prediction under covariate shift, *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)* (2019), arXiv:1904.06019.
 - [19] R. Vinuesa and S. L. Brunton, Enhancing computational fluid dynamics with machine learning, *Nature Computational Science* **2**, 358 (2022), arXiv:2110.02085.
 - [20] P. R. Spalart and S. R. Allmaras, A one-equation turbulence model for aerodynamic flows, in *30th Aerospace Sciences Meeting and Exhibit, AIAA-92-0439* (1992).
 - [21] F. R. Menter, Two-equation eddy-viscosity turbulence models for engineering applications, *AIAA Journal* **32**, 1598 (1994).
 - [22] I. B. Celik, U. Ghia, P. J. Roache, C. J. Freitas, H. Coleman, and P. E. Raad, Procedure for estimation and reporting of uncertainty due to discretization in CFD applications, *Journal of Fluids Engineering* **130**, 078001 (2008).
 - [23] P. J. Roache, Perspective: A method for uniform reporting of grid refinement studies, *Journal of Fluids Engineering* **116**, 405 (1994).